

Supporting Materials for “Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes”

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S1 Proof of Theorem 1 (DMOP- α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x_{1:n,j}^{A,P,\theta}$ contains the history of the prediction particle $x_{n,j}^{P,\theta}$ at observation times $1:n$. Similarly, $g_{i,j}^{A,P,\theta}$ is the ancestral weight of $x_{n,j}^{P,\theta}$ at observation time i .

Lemma S1. Write $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$ for the DMOP- α gradient estimate (or, equivalently, the gradient of MOP- α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta), \quad (\text{S1})$$

yielding the bootstrap filter estimator of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider the case of MOP- α when $\alpha = 1$ and $\theta = \phi$. We then have a nice telescoping product property for the after-resampling likelihood estimate:

$$\begin{aligned} \hat{\mathcal{L}}^1(\theta) &:= \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n-1,j}^{F,\theta}} \\ &= \left(\frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}, \end{aligned} \quad (\text{S2})$$

where the third equality follows from the choice of $\alpha = 1$, and the fourth equality is the resulting telescoping property. The log-derivative identity lets us decompose the score estimate as

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{\nabla_{\theta} \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{\nabla_{\theta} \left(\frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}}{\prod_{n=1}^N L_n^{\phi}} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}. \quad (\text{S3})$$

From (S3), we see that the derivative of the log-likelihood estimate is

$$\nabla_{\theta} \hat{\ell}^1(\theta) := \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}. \quad (\text{S4})$$

We proceed to decompose (S4). First, observe that as $\alpha = 1$,

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta} \frac{g_{n,j}^{\theta}}{g_{n,j}^{\phi}} = \prod_{i=1}^n \frac{g_{i,j}^{A,P,\theta}}{g_{i,j}^{A,P,\phi}}, \quad (\text{S5})$$

Note that the right hand side of (S5) is the cumulative product of measurement density ratios over the ancestral trajectory of the j -th prediction particle at timestep n . We then use the log-derivative identity again, yielding the following expression for the gradient of the log-weights as the sum of the log measurement densities over the ancestral trajectory,

$$\frac{\nabla_{\theta} w_{n,j}^{P,\theta}}{w_{n,j}^{P,\theta}} = \nabla_{\theta} \log w_{n,j}^{P,\theta} = \nabla_{\theta} \log \left(\prod_{i=1}^n \frac{g_{i,j}^{A,P,\theta}}{g_{i,j}^{A,P,\phi}} \right) \quad (\text{S6})$$

$$= \nabla_{\theta} \sum_{i=1}^n \left(\log g_{i,j}^{A,P,\theta} - \log g_{i,j}^{A,P,\phi} \right) \quad (\text{S7})$$

$$= \sum_{i=1}^n \nabla_{\theta} \log g_{i,j}^{A,P,\theta}. \quad (\text{S8})$$

This is equal to the gradient of the logarithm of the conditional density of the observed measurements given the ancestral trajectory of the j -th prediction particle up to timestep n ,

$$\nabla_{\theta} \sum_{n=1}^N \log g_{n,j}^{A,P,\theta} = \nabla_{\theta} \log \left(\prod_{n=1}^N g_{n,j}^{A,P,\theta} \right) \quad (\text{S9})$$

$$= \nabla_{\theta} \log \left(\prod_{n=1}^N f_{Y_n|X_n}(y_n^* | x_{n,j}^{A,P,\theta}; \theta) \right) \quad (\text{S10})$$

$$= \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta). \quad (\text{S11})$$

Multiplying both sides of the expression by $w_{N,j}^{P,\theta}$ yields an expression for the gradient of the weights at timestep N ,

$$\nabla_{\theta} w_{N,j}^{P,\theta} = w_{N,j}^{P,\theta} \sum_{n=1}^N \nabla_{\theta} \log g_{n,j}^{A,P,\theta} = w_{N,j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta). \quad (\text{S12})$$

Substituting the above identity into the log-likelihood decomposition obtained earlier in Equation S3 yields

$$\begin{aligned} \nabla_{\theta} \hat{\ell}^1(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,k_j}^{P,\theta} \\ &= \frac{1}{J} \sum_{j=1}^J w_{N,k_j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta). \end{aligned} \quad (\text{S13})$$

Finally, observing that $\theta = \phi$ implies $w_{N,j}^{F,\theta} = 1$, we obtain

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta). \quad (\text{S14})$$

This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. $\square$

Note that the variance of the DMOP- α log-likelihood derivative estimate scales poorly with N when $\theta \neq \phi$. This can be seen by observing that

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,k_j}^{P,\theta}$$

$$= \frac{1}{J} \sum_{j=1}^J w_{N,k_j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta). \quad (\text{S15})$$

When $\theta \neq \phi$, we see that $w_{N,k_j}^{P,\theta} = O(c^N)$. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite- J upper bound.

Lemma S2. Write $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$ for the DMOP- α gradient estimate (or, equivalently, the gradient of MOP- α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,

$$\nabla_{\theta} \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S16})$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\phi}; \theta)} \quad (\text{S17})$$

as shorthand for the measurement density ratios. Observe that, when $\alpha = 0$, the likelihood estimate becomes

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \quad (\text{S18})$$

$$= \prod_{n=1}^N L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \quad (\text{S19})$$

$$= \prod_{n=1}^N L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\phi}; \theta)}. \quad (\text{S20})$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\nabla_{\theta} \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_{\theta} \log \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S21})$$

$$= \sum_{n=1}^N \frac{\nabla_{\theta} \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S22})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_{\theta} s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S23})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_{\theta} f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} \quad (\text{S24})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S25})$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^J s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. $\square$

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n , the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^{\theta}(Z_n) = f_{Y_n|X_n}(y_n^* | X_n^{\theta}; \theta), \quad W_{\theta}(Z_N) = \prod_{n=1}^N G_n^{\theta}(Z_n), \quad (\text{S26})$$

$$\psi_{\theta}(Z_N) = \nabla_{\theta}^{\text{tot}} \log W_{\theta}(Z_N), \quad \Xi_{\theta}(Z_N) = \nabla_{\theta}^{2,\text{tot}} \log W_{\theta}(Z_N). \quad (\text{S27})$$

The superscript “tot” means that the derivative passes through the reparameterized simulated path while the base random numbers and the discrete resampling indices are held fixed.

Write $\rho_N^J = J^{-1} \sum_{j=1}^J \delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_{\theta}(Z_N)F(Z_N)\}}{\mathbb{E}\{W_{\theta}(Z_N)\}}. \quad (\text{S28})$$

Assumptions (A3) and (A5) give $|\partial_{\theta_i} W_{\theta}| \leq N \bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_{\theta} \ell(\theta) = \rho_N(\psi_{\theta}), \quad \nabla_{\theta} \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_{\theta}). \quad (\text{S29})$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1, \dots, N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \leq q \leq N$, write $\rho_q^J = J^{-1} \sum_{j=1}^J \delta_{Z_{q,j}}$, where the $Z_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $Z_{N,j} = Z_{N,j}$. Write $V_q^{\theta}(Z_q) = G_{q+1}^{\theta}(Z_q)$ for $q = 0, \dots, N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^{\theta}$ append the next independent simulator innovation and its gradients for $0 \leq q \leq N-2$, and put $\mathcal{M}_N^{\theta} = \text{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^{\theta}(\mu)(dz') = \frac{\int \mu(dz) V_q^{\theta}(z) \mathcal{M}_{q+1}^{\theta}(z, dz')}{\mu(V_q^{\theta})}, \quad 0 \leq q < N. \quad (\text{S30})$$

Conditionally on the current cloud, the J children are independent with common law $\Phi_{q+1}^\theta(\rho_q^J)$. The final observation is handled by the $q = N - 1$ transition, namely multinomial selection with potential $V_{N-1}^\theta = G_N^\theta$ followed by the identity mutation. Thus ρ_N^J is exactly the terminal empirical measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \leq q \leq N$, let

$$Q_{q,N}^\theta F(z_q) = \mathbb{E} \left[F(Z_N) \prod_{r=q}^{N-1} V_r^\theta(Z_r) \middle| Z_q = z_q \right], \quad (\text{S31})$$

$$P_{q,N}^\theta F = \frac{Q_{q,N}^\theta F}{Q_{q,N}^\theta 1}, \quad g_{q,N} = \sup_{z,z'} \frac{Q_{q,N}^\theta 1(z)}{Q_{q,N}^\theta 1(z')}, \quad b_{q,N} = \sup_{\text{osc}(F) \leq 1} \text{osc}(P_{q,N}^\theta F), \quad (\text{S32})$$

where $Q_{N,N}^\theta F = F$ and $\text{osc}(F) = \sup_{z,z'} |F(z) - F(z')|$. Thus $0 \leq b_{q,N} \leq 1$. With

$$R_G = \frac{\bar{G}}{\underline{G}}, \quad S_N = \sum_{q=0}^N g_{q,N}^3 b_{q,N}, \quad K_N = 32 S_N, \quad K_N^{\text{pf}} = 32 \sum_{r=0}^N R_G^{3r}, \quad (\text{S33})$$

the potential bounds give, term by term,

$$\underline{G}^{N-q} \leq Q_{q,N}^\theta 1(z_q) \leq \bar{G}^{N-q}, \quad g_{q,N} \leq R_G^{N-q}, \quad K_N \leq K_N^{\text{pf}}. \quad (\text{S34})$$

In particular, the constant used in the theorem is the closed-form quantity

$$K_N^{\text{pf}} = \begin{cases} 32(N+1), & R_G = 1, \\ 32 \frac{R_G^{3(N+1)} - 1}{R_G^3 - 1}, & R_G > 1. \end{cases} \quad (\text{S35})$$

Here is the constant calculation behind the tail bound used below. Let r_N^{DM} and β_N^{DM} denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r_N^{\text{DM}} \leq 4S_N, \quad (\beta_N^{\text{DM}})^2 \leq 4 \sum_{q=0}^N g_{q,N}^2 b_{q,N}^2 \leq 4S_N, \quad (\text{S36})$$

where the last inequality uses $g_{q,N} \geq 1$ and $0 \leq b_{q,N} \leq 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \leq \log 6$, then $6e^{-x} \geq 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \leq 2x + 2\sqrt{x}$ for $h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \leq 8x$ to obtain

$$\begin{aligned} \frac{r_N^{\text{DM}}}{J} \{1 + h^{-1}(x)\} &\leq \frac{u^2}{8x} \{1 + 2x + 2\sqrt{x}\} \leq u^2 \leq \frac{u}{2}, \\ \beta_N^{\text{DM}} \sqrt{\frac{2x}{J}} &\leq 2\sqrt{S_N} \sqrt{\frac{2x}{J}} = \frac{u}{2}. \end{aligned}$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr(|(\rho_N^J - \rho_N)(F)| > \text{osc}(F)u) \leq 6 \exp\left(-\frac{Ju^2}{K_N}\right) \leq 6 \exp\left(-\frac{Ju^2}{K_N^{\text{pf}}}\right). \quad (\text{S37})$$

for every F with $\text{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho_N^J - \rho_N)(F) = 0$ identically. Equations S33– S37 replace all unnamed model-dependent concentration constants.

Lemma S3 (Explicit concentration of the DMOP-1 gradient). *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1 \leq i \leq p} \text{osc}(\psi_{\theta,i}), \quad u_g(J, \delta) = \sqrt{\frac{K_N^{\text{pf}}}{J} \log \frac{6p}{\delta}}. \quad (\text{S38})$$

If $J \geq 4K_N^{\text{pf}} \log(6p/\delta)$, then, with probability at least $1 - \delta$,

$$\|\nabla_{\theta} \hat{\ell}_J^{A,1}(\theta) - \nabla_{\theta} \ell(\theta)\|_2 \leq A_N(\theta) \sqrt{p} u_g(J, \delta). \quad (\text{S39})$$

Moreover, Assumption (A3) gives $A_N(\theta) \leq 2NG'(\theta)$. Consequently, the explicit condition

$$J \geq K_N^{\text{pf}} \log \frac{6p}{\delta} \max \left\{ 4, \frac{4pN^2 G'(\theta)^2}{\epsilon^2} \right\} \quad (\text{S40})$$

implies that the left side of equation S39 is at most ϵ with probability at least $1 - \delta$.

Proof. The lower bound on J makes $u_g(J, \delta) \leq 1/2$. Apply equation S37 to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over the p coordinates gives

$$\max_{1 \leq i \leq p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \leq A_N(\theta) u_g(J, \delta)$$

with probability at least $1 - \delta$. Now use $\|v\|_2 \leq \sqrt{p}\|v\|_{\infty}$ and equation S29. Finally, $|\psi_{\theta,i}| \leq NG'(\theta)$, so $A_N(\theta) \leq 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta) \sqrt{p} u_g(J, \delta) \leq \epsilon$ gives equation S40. $\square$

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

Lemma S4 (Explicit concentration of the DMOP-1 particle Hessian). *Suppose in addition that Assumption (A3) gives the entrywise bound $|\Xi_{\theta,ab}(Z_N)| \leq NG''(\theta)$ for $1 \leq a, b \leq p$. Define*

$$d_p = \frac{p(p+3)}{2}, \quad H_N(\theta) = 2p\{NG''(\theta) + 3N^2 G'(\theta)^2\}, \quad u_H(J, \delta) = \sqrt{\frac{K_N^{\text{pf}}}{J} \log \frac{6d_p}{\delta}}. \quad (\text{S41})$$

If $J \geq 4K_N^{\text{pf}} \log(6d_p/\delta)$, then, with probability at least $1 - \delta$,

$$\|\nabla_{\theta}^2 \hat{\ell}_J^{A,1}(\theta) - \nabla_{\theta}^2 \ell(\theta)\|_{\text{op}} \leq H_N(\theta) u_H(J, \delta). \quad (\text{S42})$$

Fix $0 < \kappa < \gamma$. If

$$\gamma I_p \preceq -\nabla_{\theta}^2 \ell(\theta) \preceq \Gamma I_p \quad \text{and} \quad J \geq K_N^{\text{pf}} \log \frac{6d_p}{\delta} \max \left\{ 4, \frac{H_N(\theta)^2}{\kappa^2} \right\}, \quad (\text{S43})$$

then, with probability at least $1 - \delta$,

$$(\gamma - \kappa) I_p \preceq -\nabla_{\theta}^2 \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma + \kappa) I_p. \quad (\text{S44})$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_{\phi} + \log \left\{ \frac{1}{J} \sum_{j=1}^J \exp(a_j(\theta)) \right\}, \quad a_j(\theta) = \log w_{N,j}^{F,\theta}, \quad a_j(\phi) = 0,$$

where C_{ϕ} is constant in θ . Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla_{\theta}^2 \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_{\theta} + \psi_{\theta} \psi_{\theta}^T) - \rho_N^J(\psi_{\theta}) \rho_N^J(\psi_{\theta})^T. \quad (\text{S45})$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial_{\theta_a \theta_b}^2 W_{\theta}| = W_{\theta} |\Xi_{\theta,ab} + \psi_{\theta,a} \psi_{\theta,b}| \leq \bar{G}^N \{NG''(\theta) + N^2 G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E} W_{\theta}$ gives the population identity

$$\nabla_{\theta}^2 \ell = \rho_N(\Xi_{\theta} + \psi_{\theta} \psi_{\theta}^T) - \rho_N(\psi_{\theta}) \rho_N(\psi_{\theta})^T. \quad (\text{S46})$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2 G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1 , and the symmetric entries of $\Xi + \psi \psi^T$ have oscillation at most A_2 . Apply equation S37 jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\begin{aligned} \|(\rho_N^J - \rho_N)(\psi)\|_2 &\leq A_1 \sqrt{p} u_H(J, \delta), \\ \|(\rho_N^J - \rho_N)(\Xi + \psi \psi^T)\|_{\text{op}} &\leq p A_2 u_H(J, \delta). \end{aligned}$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p} NG'(\theta)$. Therefore

$$\begin{aligned} \|\rho_N^J(\psi) \rho_N^J(\psi)^T - \rho_N(\psi) \rho_N(\psi)^T\|_{\text{op}} \\ \leq 2\sqrt{p} NG'(\theta) \|(\rho_N^J - \rho_N)(\psi)\|_2 \leq 4p N^2 G'(\theta)^2 u_H(J, \delta). \end{aligned}$$

Combining the last three displays with equation S45 and equation S46 proves equation S42. Condition equation S43 makes its right side at most κ ; Weyl's eigenvalue inequality then gives equation S44. $\square$

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\tilde{B} = Q \text{diag}(\lambda_1, \dots, \lambda_p) Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q \text{diag}([\lambda_1]_c^C, \dots, [\lambda_p]_c^C) Q^T, \quad [x]_c^C = \min\{C, \max\{c, x\}\}. \quad (\text{S47})$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \leq v^T B v \leq C\|v\|_2^2, \quad C^{-1}\|v\|_2^2 \leq v^T B^{-1} v \leq c^{-1}\|v\|_2^2. \quad (\text{S48})$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma S4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under equation S43; equation S47 guarantees the chosen c, C bounds for every realization.

Proof of Theorem 2. Let τ and $\mathcal{E}_M$ be the stopping time and simultaneous gradient event defined in Theorem 2. If $\mathcal{F}_m$ is the history before the m th particle-filter call, then θ_m and $\{m \leq \min(\tau, M-1)\}$ are $\mathcal{F}_m$ -measurable. Lemma S3, with failure probability δ/M , gives

$$\begin{aligned} \Pr(\mathcal{E}_M^c \mid \theta_0) &\leq \sum_{m=0}^{M-1} \mathbb{E} [\mathbf{1}_{\{m \leq \min(\tau, M-1)\}} \Pr(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m) \mid \theta_0] \\ &\leq \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta. \end{aligned} \quad (\text{S49})$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1}g_m$. If $\|g_m\|_2 \leq \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon \leq (\sigma + 1)\epsilon. \quad (\text{S50})$$

Strong convexity implies $f(\theta) - f(\theta^*) \leq \|\nabla f(\theta)\|_2^2/(2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \leq \frac{(\sigma + 1)^2 \epsilon^2}{2\gamma}. \quad (\text{S51})$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \leq \|g_m\|_2 \leq C\|p_m\|_2, \quad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \quad (\text{S52})$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned} f(\theta_m + \eta p_m) - f(\theta_m) &\leq \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma \eta^2}{2} \|p_m\|_2^2 \\ &\leq -\eta p_m^T B_m p_m + \eta \epsilon \|p_m\|_2 + \frac{\Gamma \eta^2}{2} \|p_m\|_2^2. \end{aligned} \quad (\text{S53})$$

The choices of σ and η imply

$$\frac{C}{\sigma} \|p_m\|_2^2 \leq \frac{c(1-\beta)}{2} \|p_m\|_2^2, \quad \frac{\Gamma \eta}{2} \|p_m\|_2^2 \leq \frac{c(1-\beta)}{2} \|p_m\|_2^2.$$

Using $p_m^T B_m p_m \geq c \|p_m\|_2^2$ in equation S53 now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta\beta p_m^T B_m p_m. \quad (\text{S54})$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma + 1}{\sigma} \|g_m\|_2. \quad (\text{S55})$$

Consequently, equation S48 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C} \left(\frac{\sigma}{\sigma + 1} \right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C} \left(\frac{\sigma}{\sigma + 1} \right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \quad (\text{S56})$$

Combining equation S54 and equation S56 proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q \{f(\theta_m) - f(\theta^*)\}, \quad q = 1 - \frac{2\eta\beta\gamma}{C} \left(\frac{\sigma}{\sigma + 1} \right)^2. \quad (\text{S57})$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1 - \beta) \frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \quad \text{so} \quad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while equation S51 gives its stopping conclusion after substituting $f = -\ell$. $\square$

The result is deliberately restricted to DMOP-1. For $\alpha < 1$, the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing α close to one does not by itself provide such a bound.

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N, (\pi_n)_{n=1}^N, (\rho_n)_{n=1}^N$ be sequences of probability measures on the state space $\mathcal{X}$. This is the sequence of prediction distributions $f_{X_n|Y_{1:n-1}}$, filtering distributions $f_{X_n|Y_{1:n}}$, and posterior distributions $f_{X_{1:n}|Y_{1:n}}$ that we seek to approximate with the particle filter. For any measurable bounded functional h , we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \rightarrow [0, 1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x , and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x . If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A .

Markov kernels and measures: For any measure η , any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h d\eta = \int h(x)\eta(dx), \quad (\text{S58})$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \quad (\text{S59})$$

$$(Mh)(x) = \int M(x, dy)h(y). \quad (\text{S60})$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \quad (\text{S61})$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\text{TV}} = \sup_{\|h\|_\infty \leq 1/2} |\mu(h) - \nu(h)| = \sup_{\text{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \quad (\text{S62})$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\text{TV}}(M) = \sup_{x, y \in \mathcal{X}} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}. \quad (\text{S63})$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \rightarrow [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}(y_n^*|x_{n,j}^F; \theta) = G_n(x_{n,j}^F)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^*|\cdot)$ is the conditional density of the observed measurement at time n , where we condition on the filtering particle $x_{n,j}^F$ as an element of the state space.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0 , a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0} , the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \quad (\text{S64})$$

- The map from the prediction distribution at timestep n to timestep $n + 1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n)M_{n+1}. \quad (\text{S65})$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ \dots \circ \Phi_{k+1}. \quad (\text{S66})$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^P}, \quad (\text{S67})$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^F}. \quad (\text{S68})$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. The corresponding Markov transition kernel is then

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = (\Phi_n(\eta_{n-1}^J))^{\otimes J} = \left(\Phi_n \left(\frac{1}{J} \sum_{j=1}^J \delta_{x_{n-1,j}^P} \right) \right)^{\otimes J}, \quad (\text{S69})$$

and the composition of Markov kernels on particles from timestep n to timestep $n + k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \quad (\text{S70})$$

One may wonder why Karjalainen et al. (2023) require this process to evolve on $\mathcal{X}^J$. This is because at every timestep n , we in fact draw $X_{n,j}^P | \{X_{n-1,1:J}^P = x_{n-1,1:J}^P\} \sim \mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \eta_0^{\otimes J} \mathbf{M}_{0,n}$ for $j = 1, \dots, J$.

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP- α . That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(O(\log J)) \rfloor}, \quad (\text{S71})$$

for some ϵ dependent on $\bar{G}, \underline{G}, \bar{M}, \underline{M}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP- α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP- α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters). *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$, where we inductively define given some $x_{n-1,j}^F, w_{n-1,j}^F$ comprising the filtering measure approximation π_n^J :*

1. *The prediction particles at timestep n , $x_{n,j}^P$, are drawn from $X_{n,j}^P \sim \pi_n^J M_n$, and the prediction weights are given by $w_{n,j}^P = w_{n-1,j}^F$.*
2. *The prediction measure at timestep n is given by*

$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \quad (\text{S72})$$

3. *The particles are resampled at every timestep n , yielding indices k_j according to some arbitrary probabilities $(p_{n,j})_{j=1}^J$.*
4. *The filtering measure at timestep n is given by*

$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P g_{n,j}}{\sum_{j=1}^J w_{n,j}^P g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \quad (\text{S73})$$

5. *The prediction weights at the next timestep are given by $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}$.*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting). *A pair of random vectors (X, W) drawn from some measure g **targets** another measure π if, for any measurable and bounded functional h ,*

$$E_g[h(X) \cdot W] = E_\pi[h(X)]. \quad (\text{S74})$$

*A particle approximation is a triangular array of pairs of random vectors $(X_j^J, W_j^J), j = 1, 2, \dots, J$. The particle approximation **targets** π if, for any measurable and bounded functional h ,*

$$\frac{\sum_{j=1}^J h(X_j^J) W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{\text{a.s.}} E_\pi[h(X)] \quad (\text{S75})$$

as $J \rightarrow \infty$.

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure). *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Now, let $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$ be a multinomial sample with indices k_j drawn from $\{(\tilde{X}_j^J, U_j^J)\}$ where $(\tilde{X}_j^J, U_j^J)$ is represented, on average, proportional to $\pi_j^J J$ times. Write*

$$(Y_j^J, V_j^J) = (\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J). \quad (\text{S76})$$

If the importance sampling weights U_j / π_j are bounded, then $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$ targets f_X .

Proof. Note that as the Y_j^J are a subsample from X_j^J , h can be a function of Y as well as it is one for X . We then expand

$$\frac{\sum_j h(Y_j^J) V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \quad (\text{S77})$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [f(X)]. \quad (\text{S78})$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} 0. \quad (\text{S79})$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \xrightarrow{a.s.} 0 \quad (\text{S80})$$

since an application of this result with $h(x) = 1$ provides almost sure convergence of the denominator in (S79). Write $g(\tilde{X}_j^J) = h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J}$. We therefore need to show that

$$\sum_j Z_j^J := \sum_j \left(g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J) \pi_j^J \right) \xrightarrow{a.s.} 0. \quad (\text{S81})$$

Because the functional h and importance sampling weights $u_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\mathbb{E}[(Z_j^J)^4] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent given the $\tilde{X}_j^J$ and U_j^J ,

$$\begin{aligned} \mathbb{E} \left[\left(\sum_j Z_j^J \right)^4 \middle| (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] &= J \mathbb{E} \left[(Z_1^J)^4 | (\tilde{X}_1^J, U_1^J) \right] \\ &\quad + 3J(J-1) \left(\mathbb{E}[(Z_j^J)^2 | (\tilde{X}_j^J, U_j^J)_{j=1}^J] \right) \end{aligned} \quad (\text{S82})$$

$$\leq C J^2, \quad (\text{S83})$$

for some $C > 0$. Taking expectations on both sides yields

$$\mathbb{E} \left[\left(\sum_j Z_j^J \right)^4 \right] \leq C J^2 \quad (\text{S84})$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P} \left(\left| \frac{1}{J} \sum_{j=1}^J Z_j^J \right| > \epsilon \right) \leq \frac{1}{\epsilon^4 J^4} \mathbb{E} \left[\left(\sum_{j=1}^J Z_j^J \right)^4 \right] \leq \frac{C}{\epsilon^4 J^2}, \quad (\text{S85})$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\sum_j h(X_{k_j}^J) \frac{u_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(X_j^J) \frac{u_j^J}{\pi_j^J} \xrightarrow{a.s.} 0. \quad (\text{S86})$$

Similarly, we also have that

$$\sum_j \frac{u_j^J}{\pi_j^J} \pi_j^J - \sum_j \frac{u_{k_j}^J}{\pi_{k_j}^J} \pi_{k_j}^J \xrightarrow{a.s.} 0, \quad (\text{S87})$$

and the result is proved. $\square$

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals). *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Also suppose that $\tilde{Z}_j^J \sim f_{Z|X}(\cdot | \tilde{X}_j^J)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z . Then, if the U_j^J are bounded, $\{(\tilde{Z}_j^J, U_j^J)\}$ targets f_Z .*

Proof. We want to show that, for any measurable bounded h ,

$$\frac{\sum_j h(\tilde{Z}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}[\mathbb{E}_{f_{Z|X}}[h(Z)|X]]. \quad (\text{S88})$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S89})$$

Let $\bar{U}_j^J = \frac{J U_j^J}{\sum_j U_j^J}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1} \sum_j h(\tilde{Z}_j^J) U_j^J}{J^{-1} \sum_j U_j^J} = \frac{J^{-1} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J}{J^{-1} \sum_j \bar{U}_j^J}. \quad (\text{S90})$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J, \quad (\text{S91})$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \right] = \lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j \mathbb{E} [h(\tilde{Z}_j^J) \bar{U}_j^J | \tilde{X}_j^J, \bar{U}_j^J] \right] \quad (\text{S92})$$

$$= \lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j \mathbb{E} [h(Z) | X = \tilde{X}_j^J] \bar{U}_j^J \right]. \quad (\text{S93})$$

Applying (S89) with $g(x) = \mathbb{E}[h(Z)|X = x]$, the average on the right hand side converges almost surely to $\mathbb{E}\{\mathbb{E}[h(Z)|X]\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)]. \quad (\text{S94})$$

□

Lemma S7 (Particle Posteriors). *Suppose that $\{(X_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Also suppose that $(X_j'^J, U_j'^J) = (X_j^J, U_j^J f_{Z|X}(z^*|X_j^J))$. Then, if $U_j^J f_{Z|X}(z^*|X_j^J)$ and $U_j^J f_{Z|X}(z^*|X_j^J)/f_Z(z^*)$ are bounded, $\{(X_j'^J, U_j'^J)\}$ targets $f_{X|Z}(\cdot|z^*)$.*

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J) \cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}[h(X)|z^*]. \quad (\text{S95})$$

We already have that for any measurable bounded g ,

$$\frac{\sum_j g(X_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S96})$$

Consider the following:

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \times \left(\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \right)^{-1}. \quad (\text{S97})$$

We will apply Equation (S96) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [h(X) f_{Z|X}(z^*|X)], \quad (\text{S98})$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [f_{Z|X}(z^*|X)] = f_Z(z^*). \quad (\text{S99})$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X} [h(X) f_{Z|X}(z^*|X)]}{f_Z(z^*)} = \mathbb{E}_{f_X} \left[\frac{h(X) f_{Z|X}(z^*|X)}{f_Z(z^*)} \right] \quad (\text{S100})$$

$$= \mathbb{E}_{f_X} \left[h(X) \frac{f_{X|Z}(X|z^*)}{f_X(X)} \right] = \mathbb{E}_{f_{X|Z}} [h(X)|z^*], \quad (\text{S101})$$

to give the desired result. $\square$

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution). *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^F, w_{n-1,j}^F)\}$ targets π_{n-1} . Then, Lemma S6 tells us that $\{(X_{n,j}^P, w_{n,j}^P)\}$ targets η_n . Lemma S7 tells us that $\{(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta)\}$ therefore targets π_n . Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^F, w_{n,j}^F) = (X_{n,k_j}^P, w_{n,k_j}^P g_{n,k_j}/p_{n,k_j}), \quad (\text{S102})$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n . $\square$

Proposition S1 (MOP-1 Targets the Filtering Distribution). *When $\alpha = 1$ or $\phi = \theta$, MOP- α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α , the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*; \theta)$. Then, Lemma S6 tells

us that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$. Lemma S7 tells us that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta} g_{n,j}^\theta)\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$. Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}) = (X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta} g_{n,k_j}^\theta / g_{n,k_j}^\phi), \quad (\text{S103})$$

with resampling weights proportional to $g_{n,j}^\phi$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$. $\square$

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting). *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1:n-1}^*; \theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^J g_{n,j}^\theta w_{n,j}^{P,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}, \quad (\text{S104})$$

or by the after-resampling estimate,

$$L_n^{A,\theta} = L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}. \quad (\text{S105})$$

where L_n^ϕ is as defined in the various algorithms.

Here, (S104) is a direct consequence of our earlier result that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$. To see (S105), we write the numerator of (S5) as

$$L_n^\phi \sum_{j=1}^J \left[\frac{g_{n,j}^\theta}{g_{n,j}^\phi} w_{n,j}^{P,\theta} \right] \frac{g_{n,j}^\phi}{L_n^\phi} = L_n^\phi \sum_{j=1}^J w_{n,j}^{FC,\theta} \frac{g_{n,j}^\phi}{L_n^\phi}. \quad (\text{S106})$$

Using Lemma S5, we resample according to probabilities $\frac{g_{n,j}^\phi}{L_n^\phi}$ to see this is properly estimated by

$$L_n^\phi \sum_{j=1}^J w_{n,j}^{F,\theta}, \quad (\text{S107})$$

from which we obtain (S105). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^N \left(L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right). \quad (\text{S108})$$

Since $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^J w_{0,j}^{P,\theta} = J$ on the denominator and $\sum_{j=1}^J w_{N,j}^{F,\theta}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now provide a result showing that under sufficient regularity conditions, one can interchange the order of differentiation and expectation of functionals of off-parameter resampled particle estimates, showing that if an off-parameter resampled particle estimate is consistent for some estimand, its derivative is consistent for the derivative of the estimand as well.

One may wonder why this may be necessary, given that we have already shown strong consistency for measurable bounded functionals in Theorem S1. If we require the gradient to be bounded as well, then the gradient is also a bounded functional. The answer is that the strong consistency is for expectations under the filtering distribution, so Theorem S1 only establishes strong consistency of the gradient of the estimate to its expectation under the filtering distribution, $\nabla_\theta \eta_n^J(h_\theta) \xrightarrow{a.s.} \eta_n(\nabla_\theta h_\theta)$, which is not a-priori the gradient of the estimand $\nabla_\theta \eta_n(h_\theta)$. We require an interchange of the derivative and expectation, which we show is possible when the particle estimates have two continuous uniformly bounded derivatives over all J below.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals). *Let $h_\theta : \mathcal{X} \rightarrow \mathbb{R}$ be a measurable bounded functional of particles, where $\eta_n^J(h_\theta)$ has two continuous derivatives uniformly bounded over all J by H^* for almost every $\omega \in \Omega$ and $\theta \in \Theta$. If it holds that $\eta_n^J(h_\theta) \xrightarrow{a.s.} \eta(h_\theta) = h_\theta^*$, then we also have that $\nabla_\theta \eta_n^J(h_\theta) \xrightarrow{a.s.} \eta_n(\nabla_\theta h_\theta) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*$.*

Proof. Fix $\omega \in \Omega$. The sequence $(\nabla_\theta \eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$ is uniformly bounded over all J , by assumption. The sequence is also uniformly equicontinuous. To see this, by assumption, the second derivative of $\eta_n^J(h_\theta)(\omega)|_{\theta=\theta'}$ is also uniformly bounded over all J for almost every $\omega \in \Omega$ and every $\theta' \in \Theta$. A set of functions with derivatives bounded by the same constant is uniformly Lipschitz, and therefore uniformly equicontinuous. So the sequence $(\eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$ is uniformly equicontinuous over θ for almost every $\omega \in \Omega$. Explicitly, for almost every $\omega \in \Omega$ and every $\epsilon > 0$, there exists some $\delta(\omega) > 0$ such that for every $\|\theta - \theta'\|_\infty < \delta$ and every $J \in \mathbb{N}$ we have that

$$\|\eta_n^J(h_\theta)(\omega) - \eta_n^J(h_{\theta'})(\omega)\|_\infty < \epsilon. \quad (\text{S109})$$

Then, by Arzela-Ascoli, there exists a uniformly convergent subsequence. We claim that there is only one subsequential limit. When the gradient is bounded, we can treat the gradient as a bounded functional. So by Theorem S1 the sequence $(\eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$ converges pointwise for $\theta = \phi$ and almost every $\omega \in \Omega$, and there is therefore only one subsequential limit. The sequence therefore converges uniformly to its limit $\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega)$. Therefore, with uniform convergence for the derivatives established, we can swap the limit and derivative, and obtain that for almost every $\omega \in \Omega$,

$$\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega). \quad (\text{S110})$$

Again from Theorem S1, we know that

$$\eta_n^J(h_\theta)(\omega) \rightarrow \eta_n(h_\theta) = h_\theta^* \quad (\text{S111})$$

for almost every $\omega \in \Omega$. We then have that for almost every $\omega \in \Omega$,

$$\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*, \quad (\text{S112})$$

as we wanted. $\square$

Theorem 4 is proved as a corollary of Theorem S2, where we apply $\eta_n^J(h_\theta) = \hat{\mathcal{L}}_J^1(\theta)$, $\eta_n(h_\theta) = \mathcal{L}(\theta) = h_\theta^*$, and then use the continuous mapping theorem.

S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP- α . The earlier truncation argument is retained below for comparison. The current DMOP variables are the particle location and the accumulated weight gradient. Forgetting of the location alone does not immediately control the accumulated genealogical score. We prove the DMOP-0 result below and retain the former extension to $\alpha > 0$ inactive for comparison. Section S7 gives an exact finite- J genealogy audit, the bounds that remain valid for every α , and the additional centered estimate needed for a sharper variance theorem.

First, we note the following relation between the Dobrushin contraction coefficient and the alpha-mixing coefficients in our context below.

Lemma S9. *Setting X to be the particle collection at time m , and Y at time n , we have that the alpha mixing coefficients,*

$$\int |f_{XY}(x, y) - f_X(x) f_Y(y)| dx dy < \alpha, \quad (\text{S113})$$

are bounded by the Dobrushin coefficient, i.e we have

$$\int |f_{Y|X}(y|x_1) - f_{Y|X}(y|x_2)| dy < \alpha \quad (\text{S114})$$

for all x_1, x_2 .

Proof. We rewrite the alpha-mixing assertion as

$$\int \int |f_{Y|X}(y|x) - f_Y(y)| dy f_X(x) dx. \quad (\text{S115})$$

We claim that the Dobrushin coefficient implies

$$\int |f_{Y|X}(y|x_1) - f_Y(y)| dy < \alpha \quad (\text{S116})$$

for all x_1 . This is shown as follows:

$$\int |f_{Y|X}(y|x) - f_Y(y)| dy = \int \left| \int [f_{Y|X}(y|x_1) - f_{Y|X}(y|x)] f_X(x) dx \right| dy \quad (\text{S117})$$

$$< \int \int |f_{Y|X}(y|x_1) - f_{Y|X}(y|x)| dy f_X(x) dx \quad (\text{S118})$$

$$< \int \alpha f_X(x) dx \quad (\text{S119})$$

$$= \alpha \quad (\text{S120})$$

We then have the desired result. $\square$

As a warm-up, we consider a variance bound for DMOP-0. Note that we made Assumptions (A2) and (A3) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. This is used below to recover the $O(N/J)$ variance bound for DMOP-0. The former truncation extension to $\alpha > 0$ is retained inactive for comparison. Section S7 gives the exact genealogical audit showing why the current-filter mixing argument cannot be transferred to the accumulated score functional. We start by decomposing the MOP- α estimator as follows, for the case $\theta = \phi$,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} = \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J (w_{n-1,j}^{F,\theta})^\alpha}. \quad (\text{S121})$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when $\alpha = 0$. We now prove the variance bound for this case, presented below. To our knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

Theorem S3 (Variance of MOP-0 Gradient Estimate). *The variance of DMOP-0, i.e. the algorithm of Naesseth et al. (2018), is*

$$\text{Var}(\nabla_\theta \hat{\ell}^0(\theta)) \lesssim \frac{Np G'(\theta)^2}{J}. \quad (\text{S122})$$

Proof. When $\alpha = 0$, we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,0} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta,0}}{\sum_{j=1}^J w_{n,j}^{P,\theta,0}} \quad (\text{S123})$$

$$= \prod_{n=1}^N L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} = \prod_{n=1}^N L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\phi}; \phi)}. \quad (\text{S124})$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\phi}; \phi)}, \quad (\text{S125})$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S126})$$

$$= \sum_{n=1}^N \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S127})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S128})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_\theta f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\phi}; \phi)} \quad (\text{S129})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S130})$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^J s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J :

$$\nabla_{\theta} \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right). \quad (\text{S131})$$

Therefore, for a given θ_i in the parameter vector θ ,

$$\text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}^0(\theta) \right) = \frac{1}{J^2} \text{Var} \left(\sum_{n=1}^N \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \quad (\text{S132})$$

$$\begin{aligned} &= \frac{1}{J^2} \sum_{n=1}^N \text{Var} \left(\sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \\ &\quad + 2 \sum_{m < n} \text{Cov} \left(\frac{1}{J} \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_m|X_m}(y_m^*|x_{m,j}^{F,\theta}; \theta) \right), \frac{1}{J} \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \\ &= \sum_{n=1}^N \text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m < n} \text{Cov} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right). \end{aligned} \quad (\text{S133})$$

Here, we use Assumptions (A2) and (A3) that ensure strong mixing. We know from Theorem 3.1 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k -step Markov operator from timestep n and $\beta_{\text{TV}}(M) = \sup_{x,y \in E} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(c \log(J)) \rfloor}, \quad (\text{S134})$$

i.e. the mixing time of the particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}, \underline{M}, \bar{G}, \underline{G}$ in (A2) and (A3).

Lemma 2.3 of Karjalainen et al. (2023), followed by conditional multinomial resampling and the bounded Bayes-ratio step, gives the centered L^4 bound below. Theorem 3.1 of the same paper gives the displayed particle-cloud mixing coefficient. At $\alpha = 0$ these two results give

$$\begin{aligned} \left\| \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) - \mathbb{E} \left[\frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right] \right\|_{L^4} &\leq \frac{CG'(\theta)}{\sqrt{J}}, \\ a(k) &\leq (1 - \epsilon)^{\lfloor (k-1)/[c\{1+\log J\}] \rfloor}. \end{aligned} \quad (\text{S135})$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (S135) gives

$$\text{Cov} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) \leq (1 - \epsilon)^{\frac{1}{2} \lfloor \frac{n-m-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J}. \quad (\text{S136})$$

Putting it all together, we see that

$$\text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}^0(\theta) \right) = \sum_{n=1}^N \text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m < n} \text{Cov} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) \quad (\text{S137})$$

$$\leq \frac{NG'(\theta)^2}{J} + 2 \sum_{n=1}^{N-1} (N-n)(1-\epsilon)^{\frac{1}{2} \lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J} \quad (\text{S138})$$

$$\leq \frac{G'(\theta)^2}{J} \left(N + 2 \sum_{n=1}^{N-1} (N-n)(1-\epsilon)^{\frac{1}{2} \lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \right) \quad (\text{S139})$$

$$\lesssim \frac{G'(\theta)^2 N}{J}. \quad (\text{S140})$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\text{Var}(\nabla_{\theta} \hat{\ell}^0(\theta)) \lesssim \frac{NpG'(\theta)^2}{J}. \quad (\text{S141})$$

□

Note that the factor of N that pops up here is due to the use of the unnormalized gradient. If one divides the gradient estimate by $\sqrt{N}$ before usage, the variance does not depend on the horizon. If one divides by N , the error is $O(1/\sqrt{NJ})$.

The order $NpG'(\theta)^2/J$ is the verified DMOP-0 baseline. It is not asserted to be a lower bound for every model or every value of α .

Theorem S4 (MSE and variance bounds for DMOP- α). *Suppose the observations are fixed, $J \geq 2$, $\alpha \in [0, 1)$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. The bounded measurement-gradient input in Assumption (A3) is understood to be the current function $h_n(X_n)$ in Equation (S146). Then*

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \lesssim pG'(\theta)^2 \left\{ \frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \right\}, \quad (\text{S142})$$

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \left[\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right]. \quad (\text{S143})$$

The population component of the fixed-observation bias satisfies

$$\left\| \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2 \leq CN\sqrt{p}G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \quad (\text{S144})$$

If $J \geq N$, the second term in Equation (S142) is absorbed by the first, so

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \lesssim \frac{NpG'(\theta)^2}{J(1-\alpha)^2}. \quad (\text{S145})$$

Section S7 gives the complete proof and an exact constant-weight calculation that checks the N and J dependence and explains the singularity as $\alpha \uparrow 1$.

Proof. Lemma S13 proves Equation (S142). Theorem S5 proves Equation (S143) and Equation (S144). □

S7 Sharper Bias-Variance Analysis

We now establish the estimates used in Theorem S4. Write $\nabla_\theta \ell_n^\alpha(\theta)$ for the n th conditional score produced by the deterministic differentiated population recursion. At $\theta = \phi$, its $\alpha = 1$ version is the exact conditional score $\nabla_\theta \ell_n(\theta)$. The current DMOP object needed below is the pair consisting of the particle location and the gradient of its log weight. The location follows the ordinary particle transition to which Assumption (A2) applies. Along each selected lineage, the second component obeys the existing recursion

$$\nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \alpha \nabla_\theta \log w_{n-1,k_j}^{F,\theta} \Big|_{\theta=\phi} + h_n(x_{n,j}^{F,\phi}). \quad (\text{S146})$$

Here h_n is the bounded current measurement-gradient function of the particle location. This is the derivative at $\theta = \phi$ of the log-space recursion

$$\begin{aligned} \log w_{n,j}^{P,\theta} &= \alpha \log w_{n-1,j}^{F,\theta}, \\ \log w_{n,j}^{F,\theta} &= \alpha \log w_{n-1,k_j}^{F,\theta} + \log g_{n,k_j}^\theta - \log g_{n,k_j}^\phi. \end{aligned} \quad (\text{S147})$$

At $\theta = \phi$, the log weights are zero but their gradients need not be. We do not assume that the location-log-weight-gradient pair is mixing in total variation. Instead, the proof expands the log-weight gradient into its single-time h_n contributions, applies location forgetting to those contributions, and sums their powers of α . Throughout this section, the observations are fixed. Expectations and variances are over the simulator and particle-filter random variables, conditional on those observations.

We use Assumptions (A3) and (A5) in the precise pathwise sense already required by the differentiated MOP construction. For fixed simulator random numbers and fixed resampling indices, Assumption (A5) says that the recursively simulated state path and the composed log measurement densities are continuously differentiable in θ near ϕ , and that the joint law of the base simulator random numbers does not depend on θ . The derivative in the last clause of Assumption (A3) is the resulting total derivative: uniformly in n, j and in a neighborhood of ϕ , after enlarging $G'(\theta)$ to bound that neighborhood if necessary,

$$\left\| \nabla_\theta \log g_{n,j}^{A,F,\theta} \right\|_\infty \leq G'(\theta). \quad (\text{S148})$$

These two existing assumptions also justify differentiation under the finite-horizon simulator expectations used below. Indeed, the product rule and Equation (S148) give, pathwise,

$$\begin{aligned} & \left\| \nabla_\theta \prod_{s=1}^t G_s(X_s) \right\|_\infty \\ &= \left\| \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \nabla_\theta \log G_s(X_s) \right\|_\infty \\ &\leq \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \left\| \nabla_\theta \log G_s(X_s) \right\|_\infty \\ &\leq t \bar{G}^t G'(\theta). \end{aligned} \quad (\text{S149})$$

The final expression is deterministic and finite for each fixed t . For each coordinate of θ , the mean-value theorem therefore bounds every difference quotient about ϕ by the same quantity $t\bar{G}^t G'(\theta)$. The dominated convergence theorem gives

$$\nabla_{\theta} \mathbb{E}_U \left[\prod_{s=1}^t G_s(X_s) \right] = \mathbb{E}_U \left[\nabla_{\theta} \prod_{s=1}^t G_s(X_s) \right]. \quad (\text{S150})$$

Thus the required interchange follows from the clarified Assumptions (A3) and (A5); it is not an additional assumption. Geometric population forgetting follows from the backward-kernel contraction under Assumption (A2). Lemma S11 gives a uniform weak-error bound for a current filtering functional. Its proof is not used for a historical functional carried by the genealogy; that separate issue is audited below.

For the path functionals below, evaluated at $\theta = \phi$, let ρ_t be the posterior path measure of the particle locations. Thus, for every integrable path functional h ,

$$\rho_t(h) = \frac{\mathbb{E} [h(X_{0:t}) \prod_{s=1}^t G_s(X_s)]}{\mathbb{E} [\prod_{s=1}^t G_s(X_s)]}. \quad (\text{S151})$$

Here $G_s(X_s)$ is the existing measurement potential. The accumulated weight gradient is the second component of the current pair in Equation (S146); it is not included in the location variable X_s or in the transition density appearing in Assumption (A2).

Unrolling the existing filtered-weight recursion at $\theta = \phi$ gives

$$\nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \nabla_{\theta} \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi}. \quad (\text{S152})$$

For a complete ancestral path, write h_k , only in this section, for the measurement-score contribution

$$h_k = \nabla_{\theta} \log g_k^{A,F,\theta} \Big|_{\theta=\phi} \quad (\text{S153})$$

carried on that path. In the path-space notation of Chan et al. (2018), the notation used below means explicitly

$$m(\xi_t) = \frac{1}{J} \sum_{j=1}^J \delta_{\xi_{t,j}}, \quad m(\xi_t)(h_k) = \frac{1}{J} \sum_{j=1}^J h_k(\xi_{t,j}), \quad 1 \leq k \leq t, \quad (\text{S154})$$

where $\xi_{t,j}$ is the complete ancestral particle path carried by particle j at time t . Thus $h_k(\xi_{t,j})$ is the single time- k measurement-score contribution on that particle, not an arbitrary functional denoted by m . Equation (S151) gives its population counterpart.

We now identify the lag coefficients in the conditional score. At $\theta = \phi$, all weights equal one, so differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\begin{aligned} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{\alpha}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \end{aligned} \quad (\text{S155})$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^\alpha$. In the MOP- α recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time- n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^\theta}{g_{n,k_j}^\phi}. \quad (\text{S156})$$

Consequently, the second average in Equation (S155) is pathwise

$$\frac{1}{J} \sum_{j=1}^J \nabla_\theta \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} = \frac{\alpha}{J} \sum_{j=1}^J \nabla_\theta \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \quad (\text{S157})$$

which is the empirical average over the time- $(n-1)$ filtered genealogy. Substituting Equation (S152) into Equation (S155) and changing the index in the second sum give

$$\begin{aligned} \nabla_\theta \hat{\ell}_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\ &= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \{m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r})\}. \end{aligned} \quad (\text{S158})$$

Consequently, define

$$\begin{aligned} \Delta_{n,0}^J &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta_{n,r}^J &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \leq r < n. \end{aligned} \quad (\text{S159})$$

We now verify that the right-hand population quantities are the lag coefficients of the deterministic differentiated recursion. Equations (S150) and (S153) give, at $\theta = \phi$,

$$\begin{aligned} \nabla_\theta \log \mathbb{E}_U \left[\prod_{s=1}^n G_s(X_s) \right] &= \frac{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s) \sum_{s=1}^n h_s]}{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s)]} \\ &= \sum_{s=1}^n \rho_n(h_s). \end{aligned} \quad (\text{S160})$$

Subtracting the same identity at time $n-1$ gives

$$\begin{aligned} \nabla_\theta \ell_n(\theta) &= \sum_{s=1}^n \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\ &= \rho_n(h_n) + \sum_{r=1}^{n-1} \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\ &= \sum_{r=0}^{n-1} \Delta_{n,r}. \end{aligned} \quad (\text{S161})$$

Finally, differentiating the two population weighted averages in the deterministic MOP- α recursion and using Equation (S150) gives

$$\begin{aligned}
\nabla_{\theta} \ell_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r}) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\
&= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}.
\end{aligned} \tag{S162}$$

Lemma S10 (Lagwise first-derivative forgetting). *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0, 1)$ such that, for $0 \leq r < n$,*

$$\|\Delta_{n,r}\|_2 \leq C\sqrt{p} G'(\theta) \varrho^r. \tag{S163}$$

Proof. We derive the result from the posterior update and the contraction of the posterior backward kernels.

First suppose that $1 \leq r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \dots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of h_k in Equation (S153). By Equation (S148),

$$|h_{k,i}| \leq G'(\theta). \tag{S164}$$

Because $h_{k,i}$ is measurable with respect to the canonical extended path through time $k \leq n - 1$, Equation (S151) and conditional expectation with respect to the new base simulator variable give

$$\begin{aligned}
\rho_n(h_{k,i}) &= \frac{\mathbb{E}_U \left[h_{k,i} \prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{G_n(X_n) \mid X_{n-1}\} \right]}{\mathbb{E}_U \left[\prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{G_n(X_n) \mid X_{n-1}\} \right]} \\
&= \frac{\rho_{n-1} \{h_{k,i} M_n(G_n)\}}{\rho_{n-1} M_n(G_n)},
\end{aligned} \tag{S165}$$

where the first equality uses the independence of the new base simulator variable from the path through time $n - 1$. The second equality first uses the identity below and then divides its numerator and denominator by $\mathbb{E}_U \{\prod_{s=1}^{n-1} G_s(X_s)\}$. Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n) G_n(x_n). \tag{S166}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \geq \underline{G} \int M_n(x_{n-1}, dx_n) = \underline{G}, \tag{S167}$$

$$M_n(G_n)(x_{n-1}) \leq \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \tag{S168}$$

We next control how much information about the path through time k can remain at time $n-1$. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1} , written only in this proof as B_s , is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1})f_{X_s|X_{s-1}}(x_s | x_{s-1}; \theta)}{\int \pi_{s-1}(du)f_{X_s|X_{s-1}}(x_s | u; \theta)}. \quad (\text{S169})$$

Assumption (A2) gives the following minorization for every measurable set A :

$$\begin{aligned} B_s(x_s, A) &= \frac{\int_A \pi_{s-1}(dx_{s-1})f_{X_s|X_{s-1}}(x_s | x_{s-1}; \theta)}{\int \pi_{s-1}(du)f_{X_s|X_{s-1}}(x_s | u; \theta)} \\ &\geq \frac{\underline{M} \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{M}}{\bar{M}} \pi_{s-1}(A). \end{aligned} \quad (\text{S170})$$

Put $\epsilon = \underline{M}/\bar{M}$. If $\epsilon < 1$, Equation (S170) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1 - \epsilon) R_s(x_s, \cdot), \quad (\text{S171})$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1 - \epsilon} \quad (\text{S172})$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x'_s ,

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x'_s) &= (1 - \epsilon) \{R_s(f)(x_s) - R_s(f)(x'_s)\}, \\ |B_s(f)(x_s) - B_s(f)(x'_s)| &\leq (1 - \epsilon) \text{osc}(f). \end{aligned} \quad (\text{S173})$$

If $\epsilon = 1$, Equation (S170) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x'_s gives

$$\text{osc}\{B_s(f)\} \leq (1 - \epsilon) \text{osc}(f). \quad (\text{S174})$$

The function $h_{k,i} = h_{k,i}(X_k)$ is a bounded current function of the particle location X_k . Repeated conditioning through the backward kernels therefore gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i}(X_k) \mid X_{n-1} = x_{n-1}\} = (B_{n-1}B_{n-2} \cdots B_{k+1}h_{k,i})(x_{n-1}). \quad (\text{S175})$$

There are $n - 1 - k = r - 1$ backward kernels in this display. Moreover, Equation (S164) gives

$$\begin{aligned} \text{osc}(h_{k,i}) &\leq 2\|h_{k,i}\|_\infty \\ &\leq 2G'(\theta). \end{aligned} \quad (\text{S176})$$

If $r = 1$, Equation (S175) contains no backward kernel. If $r \geq 2$, applying Equation (S174) successively gives

$$\text{osc}[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}]$$

$$\leq 2G'(\theta) \begin{cases} 1, & r = 1, \\ (1 - \epsilon)^{r-1}, & r \geq 2. \end{cases} \quad (\text{S177})$$

To finish, abbreviate the conditional expectation on the left of Equation (S177) by H , only for the next calculation. Since the X_{n-1} -marginal of ρ_{n-1} is π_{n-1} , the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \quad (\text{S178})$$

$$\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}. \quad (\text{S179})$$

Substituting Equations (S178)– (S179) into Equation (S165) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$\begin{aligned} \Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\ &= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)} \\ &= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}. \end{aligned} \quad (\text{S180})$$

For any probability measure μ and bounded real f ,

$$|f(x) - \mu(f)| \leq \text{osc}(f). \quad (\text{S181})$$

The first positive lag also inherits one contraction from the forward transition. To see this directly, fix $x^\star$. Assumption (A2) gives, for every measurable set A ,

$$\begin{aligned} M_n(x, A) &= \int_A f_{X_n|X_{n-1}}(x_n | x; \theta) dx_n \\ &\geq \underline{M} \int_A dx_n \\ &\geq \frac{\underline{M}}{M} \int_A f_{X_n|X_{n-1}}(x_n | x^\star; \theta) dx_n \\ &= \epsilon M_n(x^\star, A). \end{aligned} \quad (\text{S182})$$

If $\epsilon < 1$, Equation (S182) permits the decomposition

$$M_n(x, \cdot) = \epsilon M_n(x^\star, \cdot) + (1 - \epsilon)R_n(x, \cdot), \quad (\text{S183})$$

where R_n is a Markov kernel. Therefore, for any x, x' ,

$$\begin{aligned} M_n(G_n)(x) - M_n(G_n)(x') &= (1 - \epsilon)\{R_n(G_n)(x) - R_n(G_n)(x')\}, \\ |M_n(G_n)(x) - M_n(G_n)(x')| &\leq (1 - \epsilon) \text{osc}(G_n) \\ &\leq (1 - \epsilon)(\bar{G} - \underline{G}). \end{aligned} \quad (\text{S184})$$

Taking the supremum over x, x' gives

$$\text{osc}\{M_n(G_n)\} \leq (1 - \epsilon)(\bar{G} - \underline{G}). \quad (\text{S185})$$

If $\epsilon = 1$, Equation (S182) implies that $M_n(x, \cdot)$ does not depend on x , so the left side of Equation (S185) is zero and the same bound holds. Taking absolute values in Equation (S180), using Equations (S181)–(S185), and then using Equation (S177), yields

$$\begin{aligned} |\Delta_{n,r,i}| &\leq \frac{\text{osc}(H) \text{osc}\{M_n(G_n)\}}{\underline{G}} \\ &\leq \frac{2(\bar{G} - \underline{G})}{\underline{G}} G'(\theta) \begin{cases} (1 - \epsilon), & r = 1, \\ (1 - \epsilon)^r, & r \geq 2. \end{cases} \end{aligned} \quad (\text{S186})$$

When $\epsilon < 1$, take

$$\varrho = 1 - \epsilon = 1 - \frac{M}{\bar{M}}. \quad (\text{S187})$$

Then the two cases in Equation (S186) both equal ϱ^r . Summing the squared coordinate bounds and taking square roots give

$$\begin{aligned} \|\Delta_{n,r}\|_2 &= \left\{ \sum_{i=1}^p |\Delta_{n,r,i}|^2 \right\}^{1/2} \\ &\leq C\sqrt{p} G'(\theta) \varrho^r, \quad 1 \leq r < n. \end{aligned} \quad (\text{S188})$$

Finally, when $r = 0$, Equation (S159) and Equation (S148) give

$$\begin{aligned} \|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\ &\leq \rho_n(\|h_n\|_2) \\ &\leq \sqrt{p} G'(\theta) \\ &\leq C\sqrt{p} G'(\theta) \varrho^0. \end{aligned} \quad (\text{S189})$$

Equations (S188) and (S189) prove Equation (S163). If $\epsilon = 1$, Equation (S180) and the zero oscillation in Equation (S185) show that $\Delta_{n,r} = 0$ for every $r \geq 1$, so any fixed $\varrho \in (0, 1)$ proves the same result. Thus ϱ depends only on the model minorization bounds and is independent of N, J, α , and any lag truncation k . In particular, the factor ϱ^r includes the contraction of the first positive lag; it is not only a tail bound. □

Lemma S11 (Uniform weak bias of current-particle averages). *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is $C < \infty$ such that, for every bounded real function f of the current particle location and every $t \geq 1$,*

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq \frac{C}{J} \text{osc}(f). \quad (\text{S190})$$

The constant is independent of t and J . Here “current” means measurable with respect to the time- t particle location to which Assumption (A2) applies. In particular, $h_t = h_t(X_t)$ is current at time t . For $s < t$, however, $h_s(\xi_{t,j})$ is selected through the ancestry and is not assumed to be a function of the current location alone. The lemma is therefore not applied directly to a historical h_s at a later time.

Proof. The predictor cloud consists of the current particle locations. A bounded function of the current location is therefore a current-state functional in the Feynman–Kac model already defined in Section S3. This statement does not turn an accumulated weight gradient or an individual ancestral contribution into a function of the current location. The current-location marginal of ρ_t is the filtering distribution, and hence

$$\rho_t(f) = \pi_t(f) = \Psi_t(\eta_t)(f). \quad (\text{S191})$$

Conditionally on the prediction cloud η_t^J , multinomial resampling draws the filtering particles independently from $\Psi_t(\eta_t^J)$. Therefore

$$\begin{aligned} \mathbb{E}\{m(\xi_t)(f) \mid \eta_t^J\} &= \frac{1}{J} \sum_{j=1}^J \mathbb{E}\{f(x_{t,j}^F) \mid \eta_t^J\} \\ &= \frac{1}{J} \sum_{j=1}^J \Psi_t(\eta_t^J)(f) \\ &= \Psi_t(\eta_t^J)(f). \end{aligned} \quad (\text{S192})$$

Take $\Phi_{t,t}$ to be the identity. For $1 \leq p \leq t$, define

$$\begin{aligned} Q_{t,t}(f)(x) &= G_t(x)f(x), \\ Q_{p,t}(f)(x) &= G_p(x)M_{p+1}\{Q_{p+1,t}(f)\}(x), \quad p < t. \end{aligned} \quad (\text{S193})$$

The definitions of $\Phi_{p,t}$ and Ψ_t give, for every probability measure μ ,

$$\Psi_t\{\Phi_{p,t}(\mu)\}(f) = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}. \quad (\text{S194})$$

For $p < t$,

$$Q_{p,t}(1)(x) = G_p(x) \int f_{X_{p+1}|X_p}(u \mid x; \theta) Q_{p+1,t}(1)(u) du. \quad (\text{S195})$$

Thus, for every x, x' ,

$$\begin{aligned} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} &\leq \frac{\bar{G}\bar{M} \int Q_{p+1,t}(1)(u) du}{\underline{G}\underline{M} \int Q_{p+1,t}(1)(u) du} \\ &= \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}}. \end{aligned} \quad (\text{S196})$$

For $p = t$, the same ratio is at most $\bar{G}/\underline{G}$. Put

$$R = \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}} \geq 1. \quad (\text{S197})$$

Then

$$\sup_{x, x'} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} \leq R, \quad 1 \leq p \leq t. \quad (\text{S198})$$

For $p \leq s < t$, the normalized forward kernel is

$$P_{s,s+1}^{(t)}(x, A) = \frac{\int_A f_{X_{s+1}|X_s}(u | x; \theta) Q_{s+1,t}(1)(u) du}{\int f_{X_{s+1}|X_s}(u | x; \theta) Q_{s+1,t}(1)(u) du}. \quad (\text{S199})$$

Define, only within this proof,

$$\nu_{s,t}(A) = \frac{\int_A Q_{s+1,t}(1)(u) du}{\int Q_{s+1,t}(1)(u) du}. \quad (\text{S200})$$

Assumption (A2) gives

$$\begin{aligned} P_{s,s+1}^{(t)}(x, A) &\geq \frac{\underline{M} \int_A Q_{s+1,t}(1)(u) du}{\bar{M} \int Q_{s+1,t}(1)(u) du} \\ &= \frac{\underline{M}}{\bar{M}} \nu_{s,t}(A). \end{aligned} \quad (\text{S201})$$

Consequently, with

$$\varrho = 1 - \frac{\underline{M}}{2\bar{M}} \in (0, 1), \quad (\text{S202})$$

the Dobrushin coefficient of every kernel in Equation (S199) is at most ϱ . The factor $G_s(x)$ cancels from the normalized conditional expectation, and hence

$$\frac{Q_{p,t}(f)}{Q_{p,t}(1)} = P_{p,p+1}^{(t)} P_{p+1,p+2}^{(t)} \cdots P_{t-1,t}^{(t)} f. \quad (\text{S203})$$

It follows that

$$\text{osc} \left\{ \frac{Q_{p,t}(f)}{Q_{p,t}(1)} \right\} \leq \varrho^{t-p} \text{osc}(f). \quad (\text{S204})$$

We next bound one complete selection–mutation update. For $p = 1$, put $\mu = \eta_1 = \pi_0 M_1$. For $p \geq 2$, condition on η_{p-1}^J and put

$$\mu = \Phi_p(\eta_{p-1}^J). \quad (\text{S205})$$

In either case, the particles forming η_p^J are conditionally independent with common law μ . Within this calculation, put

$$q = \frac{Q_{p,t}(1)}{\mu Q_{p,t}(1)}, \quad a = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}, \quad d = \frac{Q_{p,t}(f - a)}{\mu Q_{p,t}(1)}. \quad (\text{S206})$$

Then

$$\mu(q) = 1, \quad \mu(d) = 0, \quad (\text{S207})$$

and Equation (S194) gives

$$\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\mu)\}(f)$$

$$\begin{aligned}
&= \frac{\eta_p^J Q_{p,t}(f)}{\eta_p^J Q_{p,t}(1)} - a \\
&= \frac{\eta_p^J Q_{p,t}(f - a)}{\eta_p^J Q_{p,t}(1)} \\
&= \frac{\eta_p^J(d)}{\eta_p^J(q)}.
\end{aligned} \tag{S208}$$

Equation (S198) implies

$$R^{-1} \leq q(x) \leq R, \quad \eta_p^J(q) \geq R^{-1}, \quad \|q - 1\|_\infty \leq R. \tag{S209}$$

Writing

$$F = \frac{Q_{p,t}(f)}{Q_{p,t}(1)}, \tag{S210}$$

we have $a = \mu(qF)$ and $\mu(q) = 1$. Thus a is an average of F under the probability measure $q d\mu$, so

$$\inf F \leq a \leq \sup F. \tag{S211}$$

Since $d = q(F - a)$, Equations (S209), (S211), and (S204) give

$$\begin{aligned}
\|d\|_\infty &\leq R \operatorname{osc}(F) \\
&\leq R \varrho^{t-p} \operatorname{osc}(f).
\end{aligned} \tag{S212}$$

Conditional independence and $\mu(d) = 0$ give

$$\begin{aligned}
\mathbb{E}\{\eta_p^J(d) \mid \mu\} &= 0, \\
\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} &= \frac{1}{J} \mu(d^2), \\
\mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\} &= \frac{1}{J} \mu\{(q - 1)^2\}.
\end{aligned} \tag{S213}$$

Consequently,

$$\begin{aligned}
&\mathbb{E} \left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \\
&= -\mathbb{E} \left\{ \frac{\eta_p^J(d) \{ \eta_p^J(q) - 1 \}}{\eta_p^J(q)} \middle| \mu \right\}.
\end{aligned} \tag{S214}$$

Conditional Cauchy–Schwarz and Equations (S209)–(S213) yield

$$\begin{aligned}
&\left| \mathbb{E} \left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \right| \\
&\leq R \left[\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} \mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\} \right]^{1/2} \\
&= \frac{R}{J} \left[\mu(d^2) \mu\{(q - 1)^2\} \right]^{1/2}
\end{aligned}$$

$$\leq \frac{C}{J} \varrho^{t-p} \text{osc}(f). \quad (\text{S215})$$

It remains to write the exact full-update telescope. Since $\eta_t = \Phi_{1,t}(\eta_1)$,

$$\begin{aligned} & \mathbb{E} \Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f) \\ &= \mathbb{E} [\Psi_t\{\Phi_{1,t}(\eta_1^J)\}(f) - \Psi_t\{\Phi_{1,t}(\eta_1)\}(f)] \\ &+ \sum_{p=2}^t \mathbb{E} [\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\Phi_p(\eta_{p-1}^J))\}(f)]. \end{aligned} \quad (\text{S216})$$

Indeed, for $2 \leq p \leq t$,

$$\Phi_{p,t} \circ \Phi_p = \Phi_{p-1,t}, \quad (\text{S217})$$

so consecutive terms cancel. The only terms left are $\Psi_t(\eta_t^J)(f)$ and $-\Psi_t(\eta_t)(f)$.

Apply Equation (S215) to every summand in Equation (S216). Then

$$\begin{aligned} |\mathbb{E} \Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f)| &\leq \frac{C \text{osc}(f)}{J} \sum_{p=1}^t \varrho^{t-p} \\ &\leq \frac{C}{J(1-\varrho)} \text{osc}(f). \end{aligned} \quad (\text{S218})$$

Absorb $(1-\varrho)^{-1}$ into C and combine this display with Equations (S191) and (S192). $\square$

Remark 1 (Particle bias of the DMOP- α score). *Suppose $\theta = \phi$ and the model and resampling conditions of Lemma S13 hold. For $\alpha \in [0, 1)$,*

$$\left\| \mathbb{E} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) - \nabla_{\theta} \ell_n^{\alpha}(\theta) \right\|_2 \leq \frac{C \sqrt{p} G'(\theta)}{J(1-\alpha)}, \quad (\text{S219})$$

$$\left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2 \leq \frac{CN \sqrt{p} G'(\theta)}{J(1-\alpha)}. \quad (\text{S220})$$

Proof. The local perturbation calculation in the proof of Lemma S13 also gives the particle bias without applying Lemma S11 to a historical function. We write out the required sums. Fix a coordinate i and a conditional score time n . The terminal resampling error in Equation (S237) has conditional mean zero, and the first-order local fields in Equation (S319) have conditional mean zero. Hence only the quadratic ratio terms contribute to the bias.

For local sampling times t after creation of $h_{s,i}$, their total coefficient in the time- n conditional score is bounded by

$$\begin{aligned} & CG'(\theta) \sum_{t=1}^n \sum_{s=1}^t \alpha^{n-s} \varrho^{n-t} \\ &\leq CG'(\theta) \left(\sum_{a=0}^{\infty} (\alpha \varrho)^a \right) \left(\sum_{b=0}^{\infty} \alpha^b \right) \\ &= \frac{CG'(\theta)}{(1-\alpha \varrho)(1-\alpha)}. \end{aligned} \quad (\text{S221})$$

Here $a = n - t$ and $b = t - s$. For local sampling times preceding creation of $h_{s,i}$, Equation (S286) gives

$$\begin{aligned}
CG'(\theta) \sum_{t=1}^{n-1} \sum_{s=t+1}^n \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} \\
\leq CG'(\theta) \left(\sum_{a=1}^{\infty} \varrho^a \right) \left(\sum_{b=0}^{\infty} (\alpha \varrho)^b \right) \\
= \frac{CG'(\theta) \varrho}{(1 - \varrho)(1 - \alpha \varrho)}. \tag{S222}
\end{aligned}$$

Here $a = s - t$ and $b = n - s$. The empirical second-moment identities in Equation (S318) make every local quadratic ratio remainder equal to its displayed coefficient times at most C/J in absolute expectation. Since $1 - \alpha \varrho \geq 1 - \varrho$, Equations (S221) and (S222) therefore give

$$\left| \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) - \partial_{\theta_i} \ell_n^\alpha(\theta) \right| \leq \frac{CG'(\theta)}{J(1 - \alpha)}. \tag{S223}$$

Taking the Euclidean norm over the p coordinates proves Equation (S219). Finally,

$$\begin{aligned}
& \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^\alpha(\theta) - \nabla_{\theta} \ell^\alpha(\theta) \right\|_2 \\
& \leq \sum_{n=1}^N \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}_n^\alpha(\theta) - \nabla_{\theta} \ell_n^\alpha(\theta) \right\|_2 \\
& \leq \frac{CN \sqrt{p} G'(\theta)}{J(1 - \alpha)}, \tag{S224}
\end{aligned}$$

which proves Equation (S220). $\square$

Lemma S12 (Full-score summation-by-parts identities). *Suppose $\theta = \phi$ and $\alpha \in [0, 1]$. Then the particle and population scores satisfy Equations (S228) and (S229) below. Moreover, for every coordinate i and $t \leq N$,*

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \tag{S225}$$

Proof. We first record the summation-by-parts identity used again in the variance proof. Sum Equation (S158) over $n = 1, \dots, N$ and put $k = n - r$. Since the sums are finite,

$$\nabla_{\theta} \hat{\ell}^\alpha(\theta) = \sum_{k=1}^N \left[m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \right]. \tag{S226}$$

For each fixed k , collecting the coefficient of every $m(\xi_t)(h_k)$ gives

$$m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\}$$

$$= (1 - \alpha) \sum_{t=k}^{N-1} \alpha^{t-k} m(\xi_t)(h_k) + \alpha^{N-k} m(\xi_N)(h_k). \quad (\text{S227})$$

Indeed, the coefficients are $1 - \alpha$ at $t = k$, $\alpha^{t-k} - \alpha^{t+1-k} = (1 - \alpha)\alpha^{t-k}$ for $k < t < N$, and α^{N-k} at $t = N$. Interchanging the finite sums once more yields

$$\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} m(\xi_t) \left(\sum_{k=1}^t \alpha^{t-k} h_k \right) + m(\xi_N) \left(\sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S228})$$

The same algebra applied to Equation (S162) gives

$$\nabla_{\theta} \ell^{\alpha}(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} \rho_t \left(\sum_{k=1}^t \alpha^{t-k} h_k \right) + \rho_N \left(\sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S229})$$

For every coordinate i , Equation (S148) gives

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \quad (\text{S230})$$

Equation (S230) is identical to Equation (S225), completing the proof. $\square$

Lemma S13 (Particle error and variance of the DMOP- α score). *Suppose $J \geq 2$, $\theta = \phi$, $\alpha \in [0, 1)$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. The bounded measurement-gradient input is the current function $h_n(X_n)$ in Equation (S146). Then*

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 \lesssim p G'(\theta)^2 \left\{ \frac{N}{J(1 - \alpha)^2} + \frac{N^2}{J^2(1 - \alpha)^2} \right\}, \quad (\text{S231})$$

and

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \lesssim p G'(\theta)^2 \left\{ \frac{N}{J(1 - \alpha)^2} + \frac{N^2}{J^2(1 - \alpha)^2} \right\}. \quad (\text{S232})$$

Here the multiplicative constants suppressed by $\lesssim$ depend on the fixed model bounds, but are independent of N, J , and α , and $\text{Var}(Z) = \mathbb{E} \|Z - \mathbb{E} Z\|_2^2$. In particular, if $J \geq N$, the second term is absorbed by the first and

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \lesssim \frac{N p G'(\theta)^2}{J(1 - \alpha)^2}. \quad (\text{S233})$$

Proof. Fix a coordinate i and write

$$Z_i^J = \partial_{\theta_i} \hat{\ell}^{\alpha}(\theta), \quad z_i^{\alpha} = \partial_{\theta_i} \ell^{\alpha}(\theta). \quad (\text{S234})$$

Equation (S158) and Equation (S148) give

$$\begin{aligned} |Z_i^J| &\leq \sum_{n=1}^N \left\{ G'(\theta) + 2G'(\theta) \sum_{r=1}^{n-1} \alpha^r \right\} \\ &\leq N G'(\theta) \left\{ 1 + \frac{2}{1 - \alpha} \right\} < \infty. \end{aligned} \quad (\text{S235})$$

Thus all random variables used below are square integrable.

We first separate the error created by the time- n multinomial resampling. For the prediction paths at time n , write

$$p_{n,a} = \frac{G_n(\xi_{n,a}^P)}{\sum_{b=1}^J G_n(\xi_{n,b}^P)}, \quad a = 1, \dots, J, \quad (\text{S236})$$

and put

$$R_{n,i}^J = \frac{1}{J} \sum_{j=1}^J \left[\sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi_{n,k_j}^P) - \sum_{a=1}^J p_{n,a} \sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi_{n,a}^P) \right]. \quad (\text{S237})$$

Conditionally on the prediction paths, the J summands in Equation (S237) are independent and centered. Moreover,

$$\begin{aligned} \text{osc} \left(\sum_{s=1}^n \alpha^{n-s} h_{s,i} \right) &\leq 2G'(\theta) \sum_{s=1}^n \alpha^{n-s} \\ &\leq \frac{2G'(\theta)}{1-\alpha}. \end{aligned} \quad (\text{S238})$$

It follows that

$$\begin{aligned} \mathbb{E} \{ (R_{n,i}^J)^2 \mid \xi_{n,1:J}^P \} &= \frac{1}{J} \text{Var} \left\{ \sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi_{n,k_1}^P) \mid \xi_{n,1:J}^P \right\} \\ &\leq \frac{G'(\theta)^2}{J(1-\alpha)^2}, \end{aligned} \quad (\text{S239})$$

where the last line uses $\text{Var}(Y) \leq \text{osc}(Y)^2/4$.

We next derive the local perturbation bound for the predictable part. Fix $s \leq t \leq n$ and retain the manuscript's $h_{s,i}$. Define

$$\begin{aligned} q_{t,n}^1 &= Q_{t,n}(1), \\ q_{t,n}^0 &= \begin{cases} Q_{t,n-1}(1), & t < n, \\ 1, & t = n, \end{cases} \quad F_{t,n} = \frac{q_{t,n}^1}{q_{t,n}^0}, \end{aligned} \quad (\text{S240})$$

and, for a probability measure μ on prediction paths through time t ,

$$C_{t,n}(\mu) = \frac{\mu(h_{s,i} q_{t,n}^1)}{\mu(q_{t,n}^1)} - \frac{\mu(h_{s,i} q_{t,n}^0)}{\mu(q_{t,n}^0)}. \quad (\text{S241})$$

At $t = n$, this is precisely the predictable part of the one-lag difference:

$$\begin{aligned} C_{n,n}\{m(\xi_n^P)\} &= \sum_{a=1}^J p_{n,a} h_{s,i}(\xi_{n,a}^P) - m(\xi_n^P)(h_{s,i}), \\ C_{n,n}(\eta_n) &= \begin{cases} \rho_n(h_{s,i}) - \rho_{n-1}(h_{s,i}), & s < n, \\ \rho_n(h_{n,i}) - \eta_n(h_{n,i}), & s = n. \end{cases} \end{aligned} \quad (\text{S242})$$

When several values of s occur in the same display below, write $C_{t,n}^{(s)}$ for the same functional with the displayed $h_{s,i}$; the superscript records no additional object. The conditional-score decomposition gives the exact identities

$$Z_i^J = \sum_{n=1}^N R_{n,i}^J + \sum_{n=1}^N m(\xi_n^P)(h_{n,i}) + \sum_{n=1}^N \sum_{s=1}^n \alpha^{n-s} C_{n,n}^{(s)} \{m(\xi_n^P)\}, \quad (\text{S243})$$

$$z_i^\alpha = \sum_{n=1}^N \eta_n(h_{n,i}) + \sum_{n=1}^N \sum_{s=1}^n \alpha^{n-s} C_{n,n}^{(s)}(\eta_n). \quad (\text{S244})$$

Indeed, for $s < n$, the s th summand of $R_{n,i}^J$ plus $\alpha^{n-s} C_{n,n}^{(s)} \{m(\xi_n^P)\}$ is

$$\alpha^{n-s} \{m(\xi_n)(h_{s,i}) - m(\xi_{n-1})(h_{s,i})\}, \quad (\text{S245})$$

while for $s = n$ the three terms are

$$\begin{aligned} & \left[m(\xi_n)(h_{n,i}) - \sum_{a=1}^J p_{n,a} h_{n,i}(\xi_{n,a}^P) \right] + m(\xi_n^P)(h_{n,i}) \\ & + \left[\sum_{a=1}^J p_{n,a} h_{n,i}(\xi_{n,a}^P) - m(\xi_n^P)(h_{n,i}) \right] = m(\xi_n)(h_{n,i}). \end{aligned} \quad (\text{S246})$$

This proves Equation (S243); the population identity follows by the same cancellation. The recursion for $Q_{t,n}$ and preservation of $h_{s,i}$ after time s give, for $s < t \leq n$,

$$C_{t,n} \{\Phi_t(\mu)\} = C_{t-1,n}(\mu). \quad (\text{S247})$$

Indeed, for $l \in \{0, 1\}$,

$$\begin{aligned} \frac{\Phi_t(\mu)(h_{s,i} q_{t,n}^l)}{\Phi_t(\mu)(q_{t,n}^l)} &= \frac{\mu\{G_{t-1} M_t(h_{s,i} q_{t,n}^l)\}}{\mu\{G_{t-1} M_t(q_{t,n}^l)\}} \\ &= \frac{\mu\{h_{s,i} G_{t-1} M_t(q_{t,n}^l)\}}{\mu\{G_{t-1} M_t(q_{t,n}^l)\}} \\ &= \frac{\mu(h_{s,i} q_{t-1,n}^l)}{\mu(q_{t-1,n}^l)}. \end{aligned} \quad (\text{S248})$$

Subtracting the cases $l = 1$ and $l = 0$ proves Equation (S247).

Assumptions (A2) and (A3) imply, exactly as in Equations (S198) and (S204), that

$$\begin{aligned} \sup_{x, x'} \frac{q_{t,n}^0(x)}{q_{t,n}^0(x')} &\leq C, \\ \underline{G} &\leq F_{t,n}(x) \leq \bar{G}, \\ \text{osc}(F_{t,n}) &\leq C \varrho^{n-t}. \end{aligned} \quad (\text{S249})$$

Normalize $q_{t,n}^0$ so that $\mu(q_{t,n}^0) = 1$ and write

$$\bar{\mu}(f) = \mu(q_{t,n}^0 f). \quad (\text{S250})$$

Then

$$C_{t,n}(\mu) = \frac{\bar{\mu}(h_{s,i}F_{t,n})}{\bar{\mu}(F_{t,n})} - \bar{\mu}(h_{s,i}). \quad (\text{S251})$$

For constants h_0 and F_0 in the respective ranges of $h_{s,i}$ and $F_{t,n}$,

$$\begin{aligned} & \bar{\mu}(h_{s,i}F_{t,n}) - \bar{\mu}(h_{s,i})\bar{\mu}(F_{t,n}) \\ &= \bar{\mu}\{(h_{s,i} - h_0)(F_{t,n} - F_0)\} - \bar{\mu}(h_{s,i} - h_0)\bar{\mu}(F_{t,n} - F_0). \end{aligned} \quad (\text{S252})$$

Equations (S148) and (S249) therefore give

$$|C_{t,n}(\mu)| \leq CG'(\theta)\varrho^{n-t}. \quad (\text{S253})$$

The same cancellation controls the derivative and the quadratic remainder of the normalized update. Put

$$\mathcal{R}_l(\mu) = \frac{\mu(h_{s,i}q_{t,n}^l)}{\mu(q_{t,n}^l)}, \quad l \in \{0, 1\}. \quad (\text{S254})$$

Direct subtraction of two ratios gives

$$\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) = (\nu - \mu)D_\mu \mathcal{R}_l - \frac{(\nu - \mu)(q_{t,n}^l)}{\nu(q_{t,n}^l)}(\nu - \mu)D_\mu \mathcal{R}_l, \quad (\text{S255})$$

$$D_\mu \mathcal{R}_l(x) = \frac{q_{t,n}^l(x)}{\mu(q_{t,n}^l)}\{h_{s,i}(x) - \mathcal{R}_l(\mu)\}. \quad (\text{S256})$$

Subtracting the case $l = 0$ from $l = 1$, using $q_{t,n}^1 = q_{t,n}^0 F_{t,n}$, and applying Equation (S252) yield

$$\begin{aligned} \text{osc}\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\} &\leq C \text{osc}(h_{s,i}) \text{osc}(F_{t,n}) \\ &\leq CG'(\theta)\varrho^{n-t}. \end{aligned} \quad (\text{S257})$$

The denominators in Equation (S255) are bounded away from zero. To spell out the remainder subtraction, put

$$A_l = \frac{(\nu - \mu)(q_{t,n}^l)}{\nu(q_{t,n}^l)}, \quad B_l = (\nu - \mu)D_\mu \mathcal{R}_l, \quad l \in \{0, 1\}. \quad (\text{S258})$$

The difference of the two exact ratio remainders is

$$-A_1 B_1 + A_0 B_0 = -(A_1 - A_0)B_1 - A_0(B_1 - B_0). \quad (\text{S259})$$

For $l \in \{0, 1\}$, define

$$a_l = \frac{q_{t,n}^l}{\mu(q_{t,n}^l)}. \quad (\text{S260})$$

Then $\mu(a_l) = 1$, and direct algebra gives

$$A_l = \frac{(\nu - \mu)(a_l)}{\nu(a_l)} = 1 - \frac{1}{\nu(a_l)},$$

$$\begin{aligned}
A_1 - A_0 &= \frac{1}{\nu(a_0)} - \frac{1}{\nu(a_1)} \\
&= \frac{\nu(a_1 - a_0)}{\nu(a_0)\nu(a_1)} \\
&= \frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}.
\end{aligned} \tag{S261}$$

The normalization preceding Equation (S251) and the identity $q_{t,n}^1 = q_{t,n}^0 F_{t,n}$ give

$$a_1 - a_0 = q_{t,n}^0 \left\{ \frac{F_{t,n}}{\bar{\mu}(F_{t,n})} - 1 \right\}. \tag{S262}$$

Since $\bar{\mu}(F_{t,n})$ lies between the infimum and supremum of $F_{t,n}$, Equations (S249) and (S262) imply

$$\text{osc}(a_1 - a_0) \leq C \text{osc}(F_{t,n}) \leq C \varrho^{n-t}. \tag{S263}$$

The same uniform ratio bounds give $\inf_l \nu(a_l) \geq C^{-1}$ and $\text{osc}(a_0) \leq C$. Moreover,

$$\begin{aligned}
B_1 - B_0 &= (\nu - \mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}, \\
B_1 &= (\nu - \mu)D_\mu \mathcal{R}_1,
\end{aligned} \tag{S264}$$

where Equations (S256) and (S257) give

$$\begin{aligned}
\text{osc}(D_\mu \mathcal{R}_1) &\leq C G'(\theta), \\
\text{osc}\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\} &\leq C G'(\theta) \varrho^{n-t}.
\end{aligned} \tag{S265}$$

Substitution into Equation (S259) now gives, without discarding either empirical factor,

$$\begin{aligned}
|-A_1 B_1 + A_0 B_0| &\leq C |(\nu - \mu)(a_1 - a_0)| |(\nu - \mu)D_\mu \mathcal{R}_1| \\
&\quad + C |(\nu - \mu)(a_0)| |(\nu - \mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}|.
\end{aligned} \tag{S266}$$

Centering and dividing each function in Equation (S266) by its displayed positive oscillation gives the following bound. If a displayed oscillation is zero, its corresponding empirical integral is identically zero and that product vanishes.

$$\begin{aligned}
&|C_{t,n}(\nu) - C_{t,n}(\mu) - (\nu - \mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}| \\
&\leq C G'(\theta) \varrho^{n-t} \sum_{r=1}^2 |(\nu - \mu)(f_{1,r})| |(\nu - \mu)(f_{2,r})|,
\end{aligned} \tag{S267}$$

where all $f_{1,r}$ and $f_{2,r}$ have oscillations at most one; their additive centering constants do not change any $(\nu - \mu)$ integral. The constants depend only on Assumptions (A2) and (A3). This is the specialization of Equations (3.1)–(3.4) and Appendix A.3 of Del Moral and Rio (2011) to the difference of the two ratios in Equation (S241).

We now write the full local telescope before summing any norms. Put

$$\eta_t^J = m(\xi_t^P), \quad \mu_t^J = \begin{cases} \eta_1, & t = 1, \\ \Phi_t(\eta_{t-1}^J), & t \geq 2. \end{cases} \tag{S268}$$

In this display, Φ_t denotes the canonical path-space lift of the same predictor update: the potential is evaluated at the last location, multinomial selection chooses a complete path, and M_t appends the new location. Its last-coordinate marginal is the predictor update already defined in Section S3. Thus, conditionally on the preceding prediction cloud, the J particles forming η_t^J are independent with common law μ_t^J . For the current predictor term in Equation (S243), define

$$H_{t,n}(\mu) = \Phi_{t,n}(\mu)(h_{n,i}), \quad 1 \leq t \leq n, \quad (\text{S269})$$

where $\Phi_{n,n}$ is the identity. For $t < s$, extend the already defined continuation functional by

$$C_{t,n}^{(s)}(\mu) = C_{s,n}^{(s)}\{\Phi_{t,s}(\mu)\}. \quad (\text{S270})$$

Equations (S247) and (S217) then give the exact telescopes

$$\begin{aligned} H_{n,n}(\eta_n^J) - H_{n,n}(\eta_n) \\ = \sum_{t=1}^n [H_{t,n}(\eta_t^J) - H_{t,n}(\mu_t^J)], \end{aligned} \quad (\text{S271})$$

$$\begin{aligned} C_{n,n}^{(s)}(\eta_n^J) - C_{n,n}^{(s)}(\eta_n) \\ = \sum_{t=1}^n [C_{t,n}^{(s)}(\eta_t^J) - C_{t,n}^{(s)}(\mu_t^J)]. \end{aligned} \quad (\text{S272})$$

For example, the second term in the t th summand of Equation (S272) is $C_{t-1,n}^{(s)}(\eta_{t-1}^J)$ for $t \geq 2$, so consecutive terms cancel. At $t = 1$, $\mu_1^J = \eta_1$, and the remaining endpoints are exactly those displayed on the left. The same verification proves Equation (S271).

We now derive the required future-current bounds from Assumption (A2); no contraction condition on the pair in Equation (S146) is needed. For $t \leq n$, define the predictor semigroup

$$\tilde{Q}_{t,n}(f) = \begin{cases} f, & t = n, \\ Q_{t,n-1}\{M_n(f)\}, & t < n. \end{cases} \quad (\text{S273})$$

Thus, directly from the definition of $\Phi_{t,n}$,

$$\Phi_{t,n}(\mu)(f) = \frac{\mu \tilde{Q}_{t,n}(f)}{\mu \tilde{Q}_{t,n}(1)}. \quad (\text{S274})$$

For $t \leq r < n$, put

$$\tilde{P}_{r,r+1}^{(n)}(x, A) = \frac{\int_A M_{r+1}(x, du) \tilde{Q}_{r+1,n}(1)(u)}{\int M_{r+1}(x, du) \tilde{Q}_{r+1,n}(1)(u)}, \quad (\text{S275})$$

and

$$\tilde{P}_{t,n}^{(n)} = \tilde{P}_{t,t+1}^{(n)} \tilde{P}_{t+1,t+2}^{(n)} \cdots \tilde{P}_{n-1,n}^{(n)}, \quad \tilde{P}_{n,n}^{(n)} = I. \quad (\text{S276})$$

The same calculation as Equation (S201), now with $\tilde{Q}_{r+1,n}(1)$ in place of $Q_{r+1,n}(1)$, gives

$$\tilde{P}_{r,r+1}^{(n)}(x, A) \geq \frac{\underline{M}}{\bar{M}} \frac{\int_A \tilde{Q}_{r+1,n}(1)(u), du}{\int \tilde{Q}_{r+1,n}(1)(u), du}. \quad (\text{S277})$$

Consequently, every kernel in Equation (S275) has Dobrushin coefficient at most ϱ , and

$$\frac{\tilde{Q}_{t,n}(f)}{\tilde{Q}_{t,n}(1)} = \tilde{P}_{t,n}^{(n)} f. \quad (\text{S278})$$

It follows that

$$\begin{aligned} \text{osc}\{\tilde{P}_{t,n}^{(n)} h_{n,i}\} &\leq \varrho^{n-t} \text{osc}(h_{n,i}) \\ &\leq 2G'(\theta) \varrho^{n-t}. \end{aligned} \quad (\text{S279})$$

The first variation of $H_{t,n}$ is

$$D_\mu H_{t,n}(x) = \frac{\tilde{Q}_{t,n}(1)(x)}{\mu \tilde{Q}_{t,n}(1)} \left\{ \tilde{P}_{t,n}^{(n)} h_{n,i}(x) - H_{t,n}(\mu) \right\}. \quad (\text{S280})$$

Indeed, multiplying the right side by $(\nu - \mu)$ and using $(\nu - \mu)(1) = 0$ gives the linear term obtained by direct subtraction of $\nu \tilde{Q}_{t,n}(h_{n,i})/\nu \tilde{Q}_{t,n}(1)$ and $\mu \tilde{Q}_{t,n}(h_{n,i})/\mu \tilde{Q}_{t,n}(1)$. Since $\tilde{Q}_{t,n}(1) = Q_{t,n-1}(1)$ for $t < n$, Equation (S198) gives

$$0 < \frac{\tilde{Q}_{t,n}(1)(x)}{\mu \tilde{Q}_{t,n}(1)} \leq C. \quad (\text{S281})$$

The expression in braces in Equation (S280) is centered inside the range of $\tilde{P}_{t,n}^{(n)} h_{n,i}$. Therefore Equations (S279) and (S281) imply

$$\text{osc}\{D_\mu H_{t,n}\} \leq CG'(\theta) \varrho^{n-t}. \quad (\text{S282})$$

For $t < s \leq n$, put $\mu_s = \Phi_{t,s}(\mu)$ and

$$d_{s,n} = D_{\mu_s} C_{s,n}^{(s)}. \quad (\text{S283})$$

Because $h_{s,i} = h_{s,i}(X_s)$ is current and $q_{s,n}^0, q_{s,n}^1$ are functions of the current location at time s , Equation (S256) shows that $d_{s,n}$ is also a function of that current location. Equation (S257), applied at time s , gives

$$\text{osc}(d_{s,n}) \leq CG'(\theta) \varrho^{n-s}. \quad (\text{S284})$$

Differentiating the composition in Equation (S270) gives

$$D_\mu C_{t,n}^{(s)}(x) = \frac{\tilde{Q}_{t,s}(1)(x)}{\mu \tilde{Q}_{t,s}(1)} \left\{ \tilde{P}_{t,s}^{(s)} d_{s,n}(x) - \mu_s(d_{s,n}) \right\}. \quad (\text{S285})$$

Applying the Dobrushin contraction obtained from Equations (S277) and (S276) to $d_{s,n}$, and then using Equation (S284), gives

$$\begin{aligned} \text{osc}\{\tilde{P}_{t,s}^{(s)} d_{s,n}\} &\leq \varrho^{s-t} \text{osc}(d_{s,n}) \\ &\leq CG'(\theta) \varrho^{s-t} \varrho^{n-s} \\ &= CG'(\theta) \varrho^{n-t}. \end{aligned} \quad (\text{S286})$$

Using the normalizer bound once more in Equation (S285) proves

$$\begin{aligned} \text{osc}\{D_\mu H_{t,n}\} &\leq CG'(\theta)\varrho^{n-t}, \\ \text{osc}\{D_\mu C_{t,n}^{(s)}\} &\leq CG'(\theta)\varrho^{n-t}, \quad t < s \leq n. \end{aligned} \quad (\text{S287})$$

For the corresponding quadratic remainders, apply the exact ratio identity in Equation (S255) to $H_{t,n}$. In detail, put

$$a_H = \frac{\tilde{Q}_{t,n}(1)}{\mu\tilde{Q}_{t,n}(1)}, \quad A_H = \frac{(\nu - \mu)(a_H)}{\nu(a_H)}, \quad B_H = (\nu - \mu)D_\mu H_{t,n}. \quad (\text{S288})$$

Direct subtraction of the two ratios gives

$$H_{t,n}(\nu) - H_{t,n}(\mu) - (\nu - \mu)D_\mu H_{t,n} = -A_H B_H. \quad (\text{S289})$$

The uniform normalizer ratio and Equation (S282) give

$$\begin{aligned} \text{osc}(a_H) &\leq C, \\ \text{osc}(D_\mu H_{t,n}) &\leq CG'(\theta)\varrho^{n-t}. \end{aligned} \quad (\text{S290})$$

Since $\mu(a_H) = 1$ and $\inf \nu(a_H) \geq C^{-1}$,

$$\begin{aligned} |H_{t,n}(\nu) - H_{t,n}(\mu) - (\nu - \mu)D_\mu H_{t,n}| \\ \leq C|(\nu - \mu)(a_H)||(\nu - \mu)D_\mu H_{t,n}|. \end{aligned} \quad (\text{S291})$$

Thus this quadratic product has coefficient $CG'(\theta)\varrho^{n-t}$ after its two empirical factors are normalized by their oscillations.

For $t < s$, the propagated ratios are, explicitly,

$$\mathcal{R}_{l,t,s}(\mu) = \frac{\mu\tilde{Q}_{t,s}(h_{s,i}q_{s,n}^l)}{\mu\tilde{Q}_{t,s}(q_{s,n}^l)}, \quad l \in \{0, 1\}, \quad (\text{S292})$$

and

$$C_{t,n}^{(s)}(\mu) = \mathcal{R}_{1,t,s}(\mu) - \mathcal{R}_{0,t,s}(\mu). \quad (\text{S293})$$

Since $q_{s,n}^1 = q_{s,n}^0 F_{s,n}$,

$$\frac{\tilde{Q}_{t,s}(q_{s,n}^1)}{\tilde{Q}_{t,s}(q_{s,n}^0)} = \frac{\tilde{Q}_{t,s}(q_{s,n}^0 F_{s,n})}{\tilde{Q}_{t,s}(q_{s,n}^0)}. \quad (\text{S294})$$

The right side is a normalized forward kernel from t to s applied to $F_{s,n}$. Repeating the minorization calculation in Equation (S277), with the terminal function $q_{s,n}^0$, gives

$$\begin{aligned} \text{osc} \left\{ \frac{\tilde{Q}_{t,s}(q_{s,n}^1)}{\tilde{Q}_{t,s}(q_{s,n}^0)} \right\} &\leq \varrho^{s-t} \text{osc}(F_{s,n}) \\ &\leq C\varrho^{s-t}\varrho^{n-s} \\ &= C\varrho^{n-t}. \end{aligned} \quad (\text{S295})$$

Direct subtraction of the two ratios in Equation (S292) gives

$$\begin{aligned}\mathcal{R}_{l,t,s}(\nu) - \mathcal{R}_{l,t,s}(\mu) &= (\nu - \mu)D_\mu \mathcal{R}_{l,t,s} - A_{l,t,s}B_{l,t,s}, \\ A_{l,t,s} &= \frac{(\nu - \mu)\tilde{Q}_{t,s}(q_{s,n}^l)}{\nu\tilde{Q}_{t,s}(q_{s,n}^l)}, \\ B_{l,t,s} &= (\nu - \mu)D_\mu \mathcal{R}_{l,t,s}.\end{aligned}\tag{S296}$$

Subtracting $l = 0$ from $l = 1$ yields the exact identity

$$\begin{aligned}C_{t,n}^{(s)}(\nu) - C_{t,n}^{(s)}(\mu) - (\nu - \mu)\{D_\mu \mathcal{R}_{1,t,s} - D_\mu \mathcal{R}_{0,t,s}\} \\ = -(A_{1,t,s} - A_{0,t,s})B_{1,t,s} - A_{0,t,s}(B_{1,t,s} - B_{0,t,s}).\end{aligned}\tag{S297}$$

The semigroup recursion gives the exact identities

$$\tilde{Q}_{t,s}(q_{s,n}^l) = q_{t,n}^l, \quad l \in \{0, 1\}.\tag{S298}$$

Therefore $A_{l,t,s} = A_l$, where A_l is the normalized denominator factor already calculated in Equations (S260)–(S261). In particular,

$$\begin{aligned}A_{1,t,s} - A_{0,t,s} &= \frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}, \\ A_{0,t,s} &= \frac{(\nu - \mu)(a_0)}{\nu(a_0)}, \\ \text{osc}(a_1 - a_0) &\leq C\varrho^{n-t}, \quad \text{osc}(a_0) \leq C, \quad \inf_l \nu(a_l) \geq C^{-1}.\end{aligned}\tag{S299}$$

Furthermore,

$$\begin{aligned}B_{1,t,s} - B_{0,t,s} &= (\nu - \mu)\{D_\mu \mathcal{R}_{1,t,s} - D_\mu \mathcal{R}_{0,t,s}\}, \\ B_{1,t,s} &= (\nu - \mu)D_\mu \mathcal{R}_{1,t,s}.\end{aligned}\tag{S300}$$

Equation (S287) and the single-ratio derivative obtained directly from Equation (S292) give

$$\begin{aligned}\text{osc}\{D_\mu \mathcal{R}_{1,t,s} - D_\mu \mathcal{R}_{0,t,s}\} &\leq CG'(\theta)\varrho^{n-t}, \\ \text{osc}(D_\mu \mathcal{R}_{1,t,s}) &\leq CG'(\theta).\end{aligned}\tag{S301}$$

Substituting Equations (S299)–(S301) into Equation (S297) gives

$$\begin{aligned}&\left| C_{t,n}^{(s)}(\nu) - C_{t,n}^{(s)}(\mu) - (\nu - \mu)\{D_\mu \mathcal{R}_{1,t,s} - D_\mu \mathcal{R}_{0,t,s}\} \right| \\ &\leq C|(\nu - \mu)(a_1 - a_0)||(\nu - \mu)D_\mu \mathcal{R}_{1,t,s}| \\ &\quad + C|(\nu - \mu)(a_0)||(\nu - \mu)\{D_\mu \mathcal{R}_{1,t,s} - D_\mu \mathcal{R}_{0,t,s}\}|.\end{aligned}\tag{S302}$$

The first product in Equation (S302) has oscillation product at most $CG'(\theta)\varrho^{n-t}$, and the second has the same bound. Centering and dividing the four functions by their displayed oscillation bounds therefore places the common coefficient $CG'(\theta)\varrho^{n-t}$ outside two products of empirical errors of functions having oscillation at most one. This is the required coefficient for the quadratic remainder

when $t < s$. Together with Equation (S267) for $s \leq t$ and the direct ratio expansion for $H_{t,n}$, all quadratic remainders therefore have the coefficients displayed in Equation (S287).

Equation (S146) shows why no additional mixing coefficient is required for the accumulated weight gradient: the previously accumulated component is multiplied by α , while every new input is a bounded function $h_s(X_s)$ of the mixing location. The proof now sums those explicit coefficients before taking norms. After collecting all score contributions at their local sampling time t , those with $s \leq t \leq n$ have total coefficient

$$\begin{aligned} CG'(\theta) \sum_{s=1}^t \sum_{n=t}^N \alpha^{n-s} \varrho^{n-t} \\ = CG'(\theta) \left(\sum_{a=0}^{t-1} \alpha^a \right) \left(\sum_{b=0}^{N-t} (\alpha \varrho)^b \right) \\ \leq \frac{CG'(\theta)}{(1-\alpha)(1-\alpha \varrho)}. \end{aligned} \quad (\text{S303})$$

Those with $t < s \leq n$ have total coefficient

$$\begin{aligned} CG'(\theta) \sum_{s=t+1}^N \sum_{n=s}^N \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} \\ \leq CG'(\theta) \left(\sum_{a=1}^{\infty} \varrho^a \right) \left(\sum_{b=0}^{\infty} (\alpha \varrho)^b \right) \\ = \frac{CG'(\theta) \varrho}{(1-\varrho)(1-\alpha \varrho)}. \end{aligned} \quad (\text{S304})$$

The current predictor terms in Equation (S271) contribute

$$CG'(\theta) \sum_{n=t}^N \varrho^{n-t} \leq \frac{CG'(\theta)}{1-\varrho}. \quad (\text{S305})$$

Since $1 - \alpha \varrho \geq 1 - \varrho$, the oscillation of the aggregated first-order local function, and the coefficient of its quadratic remainder, are at most

$$\frac{CG'(\theta)}{1-\alpha}. \quad (\text{S306})$$

The same bound can be seen directly on the current location-log-weight-gradient pair. For this calculation only, write

$$w_{t,i} = \partial_{\theta_i} \log w_t^{F,\theta} \Big|_{\theta=\phi}. \quad (\text{S307})$$

Consider two pairs $(x, w_{t,i})$ and $(x', w'_{t,i})$ at time t . For each terminal time $t+r$, couple the normalized forward location kernels in Equation (S276). Equation (S201) permits the coupling to be chosen so that, at every intermediate time $t+q$,

$$\Pr\{X_{t+q} \neq X'_{t+q}\} \leq \varrho^q. \quad (\text{S308})$$

Iteration of Equation (S146) gives

$$\begin{aligned} w_{t+r,i} &= \alpha^r w_{t,i} + \sum_{q=1}^r \alpha^{r-q} h_{t+q,i}(X_{t+q}), \\ w'_{t+r,i} &= \alpha^r w'_{t,i} + \sum_{q=1}^r \alpha^{r-q} h_{t+q,i}(X'_{t+q}). \end{aligned} \quad (\text{S309})$$

Equations (S148), (S308), and (S309) imply

$$\mathbb{E}|w_{t+r,i} - w'_{t+r,i}| \leq \alpha^r |w_{t,i} - w'_{t,i}| + 2G'(\theta) \sum_{q=1}^r \alpha^{r-q} \varrho^q. \quad (\text{S310})$$

The remaining convolution is the finite sum

$$\sum_{q=1}^r \alpha^{r-q} \varrho^q = \begin{cases} \frac{\varrho(\alpha^r - \varrho^r)}{\alpha - \varrho}, & \alpha \neq \varrho, \\ r\alpha^r, & \alpha = \varrho. \end{cases} \quad (\text{S311})$$

Hence Equation (S310) shows that the coupled location–log-weight-gradient pair forgets its initial condition at the geometric rate $\max(\alpha, \varrho)^r$ apart from the factor r when $\alpha = \varrho$. This does not assert total-variation contraction of the pair: deterministic differences in the second coordinate can remain mutually singular even while their absolute distance contracts. Let $L = N - t$. The finite identities

$$(1 - \alpha) \sum_{r=0}^{L-1} \alpha^r + \alpha^L = 1, \quad (\text{S312})$$

$$(1 - \alpha) \sum_{r=q}^{L-1} \alpha^{r-q} + \alpha^{L-q} = 1, \quad 1 \leq q \leq L, \quad (\text{S313})$$

are exactly the coefficients in the full-score summation-by-parts identity (S228). Multiply Equation (S310) by $1 - \alpha$ for $r < L$, retain coefficient one for $r = L$, and sum. Interchanging the two finite sums and using Equations (S312)–(S313) gives

$$\begin{aligned} (1 - \alpha) \sum_{r=0}^{L-1} \mathbb{E}|w_{t+r,i} - w'_{t+r,i}| + \mathbb{E}|w_{t+L,i} - w'_{t+L,i}| \\ \leq |w_{t,i} - w'_{t,i}| + 2G'(\theta) \sum_{q=1}^L \varrho^q \\ \leq \frac{2G'(\theta)}{1 - \alpha} + \frac{2G'(\theta)\varrho}{1 - \varrho} \\ \leq \frac{CG'(\theta)}{1 - \alpha}, \end{aligned} \quad (\text{S314})$$

where Equation (S152) gives $|w_{t,i} - w'_{t,i}| \leq 2G'(\theta)/(1 - \alpha)$. Thus the pair need not be total-variation mixing: location coupling and the affine contraction of its log-weight-gradient component give precisely the score-continuation bound used in Equation (S306).

We next define the two local terms rather than citing a fixed-terminal formula for their sum. For the signed empirical error $\eta_t^J - \mu_t^J$, put

$$\Gamma_{t,i}^J = (\eta_t^J - \mu_t^J) \left[\sum_{n=t}^N D_{\mu_t^J} H_{t,n} + \sum_{n=t}^N \sum_{s=1}^n \alpha^{n-s} D_{\mu_t^J} C_{t,n}^{(s)} \right], \quad (\text{S315})$$

$$\begin{aligned} B_{t,i}^J &= \sum_{n=t}^N \left[H_{t,n}(\eta_t^J) - H_{t,n}(\mu_t^J) - (\eta_t^J - \mu_t^J) D_{\mu_t^J} H_{t,n} \right] \\ &\quad + \sum_{n=t}^N \sum_{s=1}^n \alpha^{n-s} \left[C_{t,n}^{(s)}(\eta_t^J) - C_{t,n}^{(s)}(\mu_t^J) - (\eta_t^J - \mu_t^J) D_{\mu_t^J} C_{t,n}^{(s)} \right]. \end{aligned} \quad (\text{S316})$$

Substitute the telescopes Equations (S271) and (S272) into the difference of Equations (S243) and (S244), and then use Equations (S315) and (S316). Term-by-term collection gives the exact identity

$$Z_i^J - z_i^\alpha = \sum_{n=1}^N R_{n,i}^J + \sum_{t=1}^N \Gamma_{t,i}^J + \sum_{t=1}^N B_{t,i}^J, \quad (\text{S317})$$

which is the local perturbation construction in Definition 3.1 and Appendix A.3 of Del Moral and Rio (2011), specialized before the sums are taken. Conditional independence of the J local particle draws, Equations (S267) and (S306), and

$$\begin{aligned} &\mathbb{E} \left[\left\{ \frac{1}{J} \sum_{j=1}^J (f(X_j) - \mu(f)) \right\}^2 \middle| \mu \right] = \frac{\mu\{(f - \mu f)^2\}}{J} \leq \frac{\text{osc}(f)^2}{4J}, \\ &\left\| \left(\frac{1}{J} \sum_{j=1}^J \{f_1(X_j) - \mu(f_1)\} \right) \left(\frac{1}{J} \sum_{j=1}^J \{f_2(X_j) - \mu(f_2)\} \right) \right\|_{L^2|\mu} \leq \frac{C \text{osc}(f_1) \text{osc}(f_2)}{J} \end{aligned} \quad (\text{S318})$$

give

$$\begin{aligned} \mathbb{E}(\Gamma_{t,i}^J \mid \mathcal{F}_{t-1}) &= 0, \\ \mathbb{E}\{(\Gamma_{t,i}^J)^2 \mid \mathcal{F}_{t-1}\} &\leq \frac{CG'(\theta)^2}{J(1-\alpha)^2}, \\ \|B_{t,i}^J\|_{L^2} &\leq \frac{CG'(\theta)}{J(1-\alpha)}. \end{aligned} \quad (\text{S319})$$

Here $\mathcal{F}_{t-1}$ contains all particle variables preceding the time- t local draws. For completeness, if $Y_j = f(X_j) - \mu(f)$, conditional independence and $\mu(Y_j) = 0$ give

$$\begin{aligned} &\mathbb{E} \left[\left(\frac{1}{J} \sum_{j=1}^J Y_j \right)^4 \middle| \mu \right] \\ &= \frac{1}{J^4} \left\{ \sum_{j=1}^J \mu(Y_j^4) + 6 \sum_{1 \leq j < k \leq J} \mu(Y_j^2) \mu(Y_k^2) \right\} \end{aligned}$$

$$\begin{aligned}
&\leq \frac{J + 3J(J-1)}{J^4} \text{osc}(f)^4 \\
&\leq \frac{3}{J^2} \text{osc}(f)^4.
\end{aligned} \tag{S320}$$

The second inequality in Equation (S318) follows by Cauchy–Schwarz applied to two instances of Equation (S320).

Within each of the two families, the variables $R_{n,i}^J$ and $\Gamma_{t,i}^J$ are martingale differences in chronological order. The two families need not be mutually orthogonal because the selection at time n also enters the next prediction cloud. Applying $\|A + B\|_{L^2}^2 \leq 2\|A\|_{L^2}^2 + 2\|B\|_{L^2}^2$ first, and then martingale orthogonality separately within each family, gives

$$\begin{aligned}
\left\| \sum_{n=1}^N R_{n,i}^J + \sum_{t=1}^N \Gamma_{t,i}^J \right\|_{L^2}^2 &\leq 2 \sum_{n=1}^N \|R_{n,i}^J\|_{L^2}^2 + 2 \sum_{t=1}^N \|\Gamma_{t,i}^J\|_{L^2}^2 \\
&\leq \frac{CNG'(\theta)^2}{J(1-\alpha)^2}.
\end{aligned} \tag{S321}$$

For the quadratic terms, Minkowski’s inequality gives

$$\begin{aligned}
\left\| \sum_{t=1}^N B_{t,i}^J \right\|_{L^2} &\leq \sum_{t=1}^N \|B_{t,i}^J\|_{L^2} \\
&\leq \frac{CNG'(\theta)}{J(1-\alpha)}.
\end{aligned} \tag{S322}$$

Combining Equations (S317), (S321), and (S322), and using $(a+b)^2 \leq 2a^2 + 2b^2$, yields

$$\mathbb{E}|Z_i^J - z_i^\alpha|^2 \leq CG'(\theta)^2 \left\{ \frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \right\}. \tag{S323}$$

Summing over $i = 1, \dots, p$ proves Equation (S231). Since z_i^α is deterministic,

$$\text{Var}(Z_i^J) = \inf_{a \in \mathbb{R}} \mathbb{E}|Z_i^J - a|^2 \leq \mathbb{E}|Z_i^J - z_i^\alpha|^2. \tag{S324}$$

Summing this inequality over the coordinates proves Equation (S232). If $J \geq N$, then $N^2/J^2 \leq N/J$, which proves Equation (S233). $\square$

Remark 2 (Exact neutral-model check). *Suppose $J \geq 2$, $\theta = \phi$, $\alpha \in [0, 1]$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then*

$$\text{Var} \left\{ \nabla_\theta \hat{\ell}^\alpha(\theta) \right\} \leq N^2 p G'(\theta)^2. \tag{S325}$$

In general, no multiplicative constant independent of N , J , and α makes

$$\frac{pG'(\theta)^2}{J} \left[\sum_{t=1}^{N-1} (1-\alpha^t)^2 + \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 \right] \tag{S326}$$

an upper bound for the variance. In particular, there is a model satisfying the assumptions of this lemma for which

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} = \frac{\sigma^2}{J} \left[N(2 - J^{-1}) + \sum_{r=1}^{N-1} (N-r) \alpha^{2r} (1 - J^{-1})^{r+1} \right], \quad (\text{S327})$$

where $\sigma^2 > 0$. Consequently, before the geometric sum saturates at the horizon, the genealogy contribution in this example has order

$$\frac{N}{J(1 - \alpha^2)}, \quad (\text{S328})$$

whereas Equation (S326) has order $J^{-1}\{N + (1 - \alpha)^{-2}\}$.

Proof. Equation (S228) and Equation (S148) give, for every coordinate i ,

$$\begin{aligned} \left| \partial_{\theta_i} \hat{\ell}^{\alpha}(\theta) \right| &\leq G'(\theta) \left[(1 - \alpha) \sum_{t=1}^{N-1} \sum_{s=1}^t \alpha^{t-s} + \sum_{s=1}^N \alpha^{N-s} \right] \\ &= G'(\theta) \left[\sum_{t=1}^{N-1} (1 - \alpha^t) + \sum_{r=0}^{N-1} \alpha^r \right] \\ &= NG'(\theta). \end{aligned} \quad (\text{S329})$$

The finite-sum identity in the last line holds directly at $\alpha = 1$. Therefore

$$\begin{aligned} \text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} &= \sum_{i=1}^p \text{Var} \left\{ \partial_{\theta_i} \hat{\ell}^{\alpha}(\theta) \right\} \\ &\leq \sum_{i=1}^p \mathbb{E} \left| \partial_{\theta_i} \hat{\ell}^{\alpha}(\theta) \right|^2 \\ &\leq N^2 p G'(\theta)^2, \end{aligned} \quad (\text{S330})$$

which proves Equation (S325).

We now construct and calculate the counterexample. Let the state space be $[0, 1]$, let

$$X_n = U_n, \quad U_n \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1], \quad (\text{S331})$$

and let $Y_n \in \{0, 1\}$. In a neighborhood of $\phi = 0$, set

$$\Pr_{\theta}(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x - 1), \quad (\text{S332})$$

and fix $y_n^* = 1$ for every n . The transition density is identically one, the simulator is infinitely differentiable, and the two measurement probabilities and their total derivatives are bounded above and away from zero on a sufficiently small neighborhood of zero. Hence Assumptions (A2), (A3), and (A5) hold. At $\theta = \phi = 0$ the measurement weights are constant and

$$h_n = \partial_{\theta} \log \Pr_{\theta}(Y_n = 1 \mid X_n) \Big|_{\theta=0} = \frac{2X_n - 1}{2},$$

$$\mathbb{E}(h_n) = 0, \quad \sigma^2 := \mathbb{E}(h_n^2) = \frac{1}{12}. \quad (\text{S333})$$

Fix an injection time s . Only within this calculation, let $M_{s,0}$ be the empirical mean of the J fresh values of h_s before the time- s resampling, and let $M_{s,d}$ be their descendant mean after d neutral multinomial selections. Put

$$e_{s,d} = M_{s,d} - M_{s,d-1}, \quad d \geq 1. \quad (\text{S334})$$

The sequence $(M_{s,d})_{d \geq 0}$ is a martingale with respect to the successive resampling sigma-fields. Consequently its increments are orthogonal. We next compute all their variances rather than invoking an asymptotic approximation. Since a uniformly selected descendant has the same marginal distribution as its ancestor,

$$\mathbb{E} \left\{ \frac{1}{J} \sum_{j=1}^J h_{s,j}^2 \right\} = \sigma^2 \quad (\text{S335})$$

at every selection generation. Conditional multinomial sampling gives

$$\mathbb{E}(e_{s,d}^2 \mid M_{s,d-1}, h_{s,1:J}^{(d-1)}) = \frac{1}{J} \left\{ \frac{1}{J} \sum_{j=1}^J (h_{s,j}^{(d-1)})^2 - M_{s,d-1}^2 \right\}. \quad (\text{S336})$$

Writing $V_d = \text{Var}(M_{s,d})$, taking expectations in Equation (S336), and using the orthogonality of $M_{s,d-1}$ and $e_{s,d}$ yield

$$\begin{aligned} V_d - V_{d-1} &= \frac{\sigma^2 - V_{d-1}}{J}, \\ V_d &= (1 - J^{-1})V_{d-1} + \frac{\sigma^2}{J}, \quad V_0 = \frac{\sigma^2}{J}. \end{aligned} \quad (\text{S337})$$

Solving the scalar recursion gives

$$\begin{aligned} V_d &= \sigma^2 \{1 - (1 - J^{-1})^{d+1}\}, \\ \text{Var}(e_{s,d}) &= V_d - V_{d-1} = \frac{\sigma^2}{J} (1 - J^{-1})^d. \end{aligned} \quad (\text{S338})$$

Put $L = N - s$. The summation-by-parts identity shows that the full-score contribution injected at time s is

$$\begin{aligned} & (1 - \alpha) \sum_{r=0}^{L-1} \alpha^r M_{s,r+1} + \alpha^L M_{s,L+1} \\ &= M_{s,1} + \sum_{r=1}^L \alpha^r (M_{s,r+1} - M_{s,r}) \\ &= M_{s,0} + e_{s,1} + \sum_{r=1}^L \alpha^r e_{s,r+1}. \end{aligned} \quad (\text{S339})$$

The first equality is the contribution of h_s in Equation (S228); the second follows by collecting the coefficient of each $M_{s,d}$; and the third uses the definition of $e_{s,d}$. Orthogonality and Equation (S338) therefore give

$$\begin{aligned} & \text{Var} \left\{ (1 - \alpha) \sum_{r=0}^{L-1} \alpha^r M_{s,r+1} + \alpha^L M_{s,L+1} \right\} \\ &= \frac{\sigma^2}{J} \left[1 + (1 - J^{-1}) + \sum_{r=1}^L \alpha^{2r} (1 - J^{-1})^{r+1} \right]. \end{aligned} \quad (\text{S340})$$

Conditional on all resampling indices, the contributions injected at different times are linear functions of independent centered collections of fresh variables h_s . Their conditional cross-products therefore have expectation zero, and hence their unconditional covariances are zero. Summing Equation (S340) over $s = 1, \dots, N$, and observing that the summand indexed by r occurs for exactly $N - r$ values of s , proves Equation (S327).

Finally, take $N = m^2$, $J = m^3$, and $\alpha = 1 - m^{-1}$. Then

$$\begin{aligned} & \frac{\sigma^2}{J} \sum_{r=1}^{N-1} (N - r) \alpha^{2r} (1 - J^{-1})^{r+1} \longrightarrow \frac{\sigma^2}{2}, \\ & \frac{1}{J} \left[\sum_{t=1}^{N-1} (1 - \alpha^t)^2 + \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 \right] \leq \frac{2N}{J} = \frac{2}{m} \longrightarrow 0. \end{aligned} \quad (\text{S341})$$

The first limit follows by dividing the geometric sum by m and using $(1 - m^{-1})^{2\lfloor mx \rfloor} \rightarrow e^{-2x}$, while $(N - r)/N \rightarrow 1$ over the $r = O(m)$ terms carrying nonnegligible mass. Thus no multiplicative constant independent of N, J , and α can make Equation (S326) a valid upper bound. The same exact sum has order $N/\{J(1 - \alpha^2)\}$ whenever $1 \ll (1 - \alpha)^{-1} \ll \min(N, J)$, which proves Equation (S328). $\square$

Remark 3 (DMOP-0 and the fixed-lag calculation). *The DMOP-0 argument given before Theorem S4 remains valid. With the manuscript's convention that $\lesssim$ suppresses the particle-system mixing-time factor, it gives*

$$\text{Var} \left(\nabla_{\theta} \tilde{\ell}^0(\theta) \right) \lesssim \frac{N p G'(\theta)^2}{J}, \quad (\text{S342})$$

and Equation (S327) has the same order at $\alpha = 0$. Moreover, $(1 - \alpha^2)^{-1} \leq (1 - \alpha)^{-2}$ for $0 \leq \alpha < 1$. Thus the exact calculation refutes the former summation-by-parts upper bound while remaining consistent with Lemma S13; the direct conditional-score proof has the N/J order exhibited by the example for every fixed $\alpha < 1$.

Lemma S14 (Population discounting bias). *Under the conditions of Lemma S10, suppose $\theta = \phi$ and $\alpha \in [0, 1]$. Then*

$$\begin{aligned} \|\nabla_{\theta} \ell_n^{\alpha}(\theta) - \nabla_{\theta} \ell_n(\theta)\|_2 &\leq C \sqrt{p} G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r) \varrho^r \\ &\leq C \sqrt{p} G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}. \end{aligned} \quad (\text{S343})$$

Proof. Equation (S162) at α and at $\alpha = 1$ gives

$$\begin{aligned}\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} (1 - \alpha^r) \Delta_{n,r} \\ &= \sum_{r=1}^{n-1} (1 - \alpha^r) \Delta_{n,r},\end{aligned}\tag{S344}$$

because $1 - \alpha^0 = 0$. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}\|\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta)\|_2 &\leq \sum_{r=1}^{n-1} (1 - \alpha^r) \|\Delta_{n,r}\|_2 \\ &\leq C\sqrt{p} G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r) \varrho^r \\ &\leq C\sqrt{p} G'(\theta) \sum_{r=1}^{\infty} (1 - \alpha^r) \varrho^r \\ &= C\sqrt{p} G'(\theta) \left\{ \sum_{r=1}^{\infty} \varrho^r - \sum_{r=1}^{\infty} (\alpha\varrho)^r \right\} \\ &= C\sqrt{p} G'(\theta) \left\{ \frac{\varrho}{1 - \varrho} - \frac{\alpha\varrho}{1 - \alpha\varrho} \right\} \\ &= C\sqrt{p} G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}.\end{aligned}\tag{S345}$$

□

Theorem S5 (MSE decomposition and bound for DMOP- α). *Suppose the observations are fixed, $J \geq 2$, $\theta = \phi$, $\alpha \in [0, 1)$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. The bounded measurement-gradient input in Assumption (A3) is the current function $h_n(X_n)$ in Equation (S146). Then*

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim p G'(\theta)^2 \left[\frac{N}{J(1 - \alpha)^2} + \frac{N^2}{J^2(1 - \alpha)^2} + \left\{ \sum_{r=1}^{N-1} (N - r)(1 - \alpha^r) \varrho^r \right\}^2 \right].\tag{S346}$$

Moreover, the exact bias–variance decomposition is

$$\begin{aligned}\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ = \text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} + \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) + \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2,\end{aligned}\tag{S347}$$

where the population discounting term satisfies

$$\|\nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta)\|_2 \leq C\sqrt{p} G'(\theta) \sum_{r=1}^{N-1} (N - r)(1 - \alpha^r) \varrho^r$$

$$\leq CN\sqrt{p}G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \quad (\text{S348})$$

The factor $\varrho \in (0, 1)$ is the one in Lemma S10 and is independent of N, J , and α .

Proof. For a square-integrable random vector Z and a deterministic vector z , direct expansion gives

$$\mathbb{E}\|Z - z\|_2^2 = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2 + \|\mathbb{E}Z - z\|_2^2, \quad (\text{S349})$$

because

$$\mathbb{E}\left[(Z - \mathbb{E}Z)^\top(\mathbb{E}Z - z)\right] = \mathbb{E}(Z - \mathbb{E}Z)^\top(\mathbb{E}Z - z) = 0. \quad (\text{S350})$$

Apply Equation (S349) with $Z = \nabla_\theta \hat{\ell}^\alpha(\theta)$ and $z = \nabla_\theta \ell(\theta)$, and add and subtract $\nabla_\theta \ell^\alpha(\theta)$ inside the deterministic norm. This gives Equation (S347).

Lemma S13 gives the direct particle error

$$\begin{aligned} & \mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left\{ \frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \right\}. \end{aligned} \quad (\text{S351})$$

Summing the first bound in Lemma S14 over $n = 1, \dots, N$, using the triangle inequality, and interchanging the two finite sums give

$$\begin{aligned} \|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2 & \leq C\sqrt{p}G'(\theta) \sum_{n=1}^N \sum_{r=1}^{n-1} (1-\alpha^r)\varrho^r \\ & = C\sqrt{p}G'(\theta) \sum_{r=1}^{N-1} (N-r)(1-\alpha^r)\varrho^r \\ & \leq CN\sqrt{p}G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}, \end{aligned} \quad (\text{S352})$$

which is Equation (S348). Pointwise, $(a+b)^2 \leq 2a^2 + 2b^2$ and the triangle inequality give

$$\begin{aligned} & \left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\ & \leq 2\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 + 2\left\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2. \end{aligned} \quad (\text{S353})$$

Take expectations in Equation (S353) and substitute Equation (S351) and Equation (S348). This proves Equation (S346). The exact bias-variance identity was proved independently at the beginning of the argument; it does not create an additional power of N . $\square$

Remark 4 (Choice of α in the fixed-observation upper bound). *For every integer $r \geq 1$ and every $0 \leq \alpha \leq 1$, the finite geometric-sum identity and the inequality $1 - \alpha^q \leq q(1 - \alpha)$ give*

$$\begin{aligned} 0 & \leq r(1 - \alpha) - (1 - \alpha^r) \\ & = (1 - \alpha) \sum_{q=0}^{r-1} (1 - \alpha^q) \end{aligned}$$

$$\leq \frac{r(r-1)}{2}(1-\alpha)^2. \quad (\text{S354})$$

Therefore

$$\begin{aligned} 0 &\leq (1-\alpha) \sum_{r=1}^{N-1} (N-r)r\varrho^r - \sum_{r=1}^{N-1} (N-r)(1-\alpha^r)\varrho^r \\ &\leq N(1-\alpha)^2 \frac{\varrho^2}{(1-\varrho)^3}, \\ \sum_{r=1}^{N-1} (N-r)(1-\alpha^r)\varrho^r &\leq N(1-\alpha) \frac{\varrho}{(1-\varrho)^2}. \end{aligned} \quad (\text{S355})$$

These inequalities are uniform in N ; no horizon-saturation argument is used for the population term. Applying the last inequality to Theorem S5 gives the following coarsening of its fixed-observation upper bound, with fixed model factors suppressed:

$$\begin{aligned} &\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ &\lesssim pG'(\theta)^2 \left\{ \frac{1}{(1-\alpha)^2} \left(\frac{N}{J} + \frac{N^2}{J^2} \right) + N^2(1-\alpha)^2 \right\}. \end{aligned} \quad (\text{S356})$$

Differentiating the expression in braces with respect to $1-\alpha$ gives

$$\begin{aligned} 0 &= -\frac{2}{(1-\alpha)^3} \left(\frac{N}{J} + \frac{N^2}{J^2} \right) + 2N^2(1-\alpha), \\ (1-\alpha)^4 &= \frac{1}{NJ} + \frac{1}{J^2}. \end{aligned} \quad (\text{S357})$$

Thus the interior rate-balancing choice in this coarsened fixed-observation upper bound is

$$1-\alpha \asymp \left(\frac{1}{NJ} + \frac{1}{J^2} \right)^{1/4}. \quad (\text{S358})$$

At this choice, Equation (S356) has order

$$pG'(\theta)^2 N \left(\frac{N}{J} + \frac{N^2}{J^2} \right)^{1/2}. \quad (\text{S359})$$

The factor $(1-\alpha)^{-1}$ in Equation (S356) enters through the replacement of finite sums $\sum_{q=0}^m \alpha^q$ in the proof of Lemma S13. Hence the effective-memory interpretation of Equation (S358) requires $(1-\alpha)^{-1} \lesssim N$. At the choice in Equation (S358),

$$\{N(1-\alpha)\}^4 \asymp \frac{N^3}{J} + \frac{N^4}{J^2}. \quad (\text{S360})$$

Thus the condition holds throughout $J \leq N$, where Equation (S359) is of order $pG'(\theta)^2 N^2/J$, and for $N \leq J \lesssim N^3$, where it is of order $pG'(\theta)^2 N^{3/2}/\sqrt{J}$. When $J \gg N^3$, the proposed choice has $(1-\alpha)^{-1} \gg N$; the finite sums in the proof of Lemma S13 must then be retained rather than replaced by $(1-\alpha)^{-1}$, and Equation (S359) is not asserted as the optimized finite-horizon rate.

This calculation minimizes a fixed-observation norm upper bound, not the actual MSE. The actual population discount bias is the signed finite sum

$$\nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) = \sum_{n=1}^N \sum_{r=1}^{n-1} (1 - \alpha^r) \Delta_{n,r}, \quad (\text{S361})$$

whose first-order coefficient can cancel. Consequently, Equation (S359) is not a lower bound on the optimally tuned MSE and does not rule out an interior α with smaller MSE than both endpoints.

There is a different balance after averaging over observations generated from the original POMP. Under the conditions of Proposition 2, martingale orthogonality replaces the fixed-observation population contribution $N^2(1 - \alpha)^2$ by $N(1 - \alpha)^2$. Combining that result with the uniform particle term in Theorem S5 gives, up to fixed model factors, the coarsened joint data-and-particle upper bound

$$pG'(\theta_0)^2 \left\{ \frac{1}{(1 - \alpha)^2} \left(\frac{N}{J} + \frac{N^2}{J^2} \right) + N(1 - \alpha)^2 \right\}. \quad (\text{S362})$$

Its interior rate-balancing choice satisfies

$$(1 - \alpha)^4 \asymp \frac{1}{J} + \frac{N}{J^2}, \quad (\text{S363})$$

and gives order

$$pG'(\theta_0)^2 N \left(\frac{1}{J} + \frac{N}{J^2} \right)^{1/2}. \quad (\text{S364})$$

For $J \geq N$, this is $pG'(\theta_0)^2 N / \sqrt{J}$. The choice in Equation (S363) is again subject to $0 < 1 - \alpha \leq 1$ and $(1 - \alpha)^{-1} \lesssim N$; for $J \geq N$, the latter condition is $J \lesssim N^4$. Outside these conditions, the choice is clipped at the endpoint or the finite geometric sums must be used directly.

Remark 5 (The location–log-weight-gradient pair and a sharper rate). At $\theta = \phi$, the gradient of the log weight obeys the existing recursions

$$\begin{aligned} \partial_{\theta_i} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} &= \alpha \partial_{\theta_i} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \\ \partial_{\theta_i} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} &= \partial_{\theta_i} \log w_{n,k_j}^{P,\theta} \Big|_{\theta=\phi} + h_{n,i}(x_{n,k_j}^{P,\phi}). \end{aligned} \quad (\text{S365})$$

The current object is therefore the pair consisting of the particle location and the gradient of its log weight. Only the location is required to mix in total variation under Assumption (A2); the affine recursion for the second component is treated directly. Two different common shifts in the second component need not mix in total variation, but a common shift cancels from the score difference in Equation (S155).

Equations (S308)–(S314) give the precise calculation. A previously accumulated difference is multiplied by α at each step. A new difference in $h_{t+q}(X_{t+q})$ occurs only while the coupled locations differ, an event having probability at most q^q . Finally,

$$(1 - \alpha) \sum_{r=q}^{L-1} \alpha^{r-q} + \alpha^{L-q} = 1 \quad (\text{S366})$$

shows that full-score summation by parts counts each such difference once, rather than adding another factor $(1 - \alpha)^{-1}$. Consequently, the memory cost is $(1 - \alpha)^{-1}$ in local root mean square and $(1 - \alpha)^{-2}$ in the displayed MSE. The factors $(1 - \alpha\rho)^{-1}$ in Equations (S303) and (S304) retain the interaction between location mixing and discounting; there is no second independent mixing penalty.

The $N^2/\{J^2(1 - \alpha)^2\}$ term in Equation (S231) comes from quadratic self-normalized-ratio remainders. It is a direct-MSE term and may represent coherent J^{-1} particle bias across observation times; it is not an additional oscillation normalization of the unnormalized score. A sharper general finite- J variance bound with the squared-weight behavior of the constant-weight calculation would follow from the centered estimate

$$\begin{aligned} & \left\| \sum_{r=0}^{n-1} \alpha^r \left\{ (\Delta_{n,r}^J)_i - \mathbb{E} (\Delta_{n,r}^J)_i \right\} \right\|_{L^4} \\ & \leq \frac{CG'(\theta)}{\sqrt{J}} \left\{ \sum_{r=0}^{n-1} \sum_{s=0}^{n-1} \alpha^{r+s} \varrho^{|r-s|} \right\}^{1/2} \end{aligned} \quad (\text{S367})$$

together with its joint-time analogue. If those estimates hold, direct covariance summation gives

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \lesssim \frac{pG'(\theta)^2}{J} \sum_{n=1}^N \sum_{r=0}^{n-1} \sum_{s=0}^{n-1} \alpha^{r+s} \varrho^{|r-s|}. \quad (\text{S368})$$

For $0 \leq \alpha < 1$, every inner double sum satisfies

$$\begin{aligned} & \sum_{r=0}^{n-1} \sum_{s=0}^{n-1} \alpha^{r+s} \varrho^{|r-s|} \\ & = \sum_{r=0}^{n-1} \alpha^{2r} + 2 \sum_{d=1}^{n-1} (\alpha\varrho)^d \sum_{r=0}^{n-1-d} \alpha^{2r} \\ & \leq \frac{1 + \alpha\varrho}{1 - \alpha\varrho} \sum_{r=0}^{n-1} \alpha^{2r}. \end{aligned} \quad (\text{S369})$$

Reversing the remaining finite sums gives

$$\begin{aligned} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^{2r} & = \sum_{r=0}^{N-1} (N - r) \alpha^{2r} \\ & = \begin{cases} \frac{N - (N + 1)\alpha^2 + \alpha^{2N+2}}{(1 - \alpha^2)^2}, & 0 \leq \alpha < 1, \\ \frac{N(N + 1)}{2}, & \alpha = 1. \end{cases} \end{aligned} \quad (\text{S370})$$

Thus this further refinement would have order $N/\{J(1 - \alpha^2)\}$ before horizon saturation and N^2/J at $\alpha = 1$. Equations (S367)–(S368) are not claimed as consequences of Assumptions (A2)–(A5): they are the precise variance-sensitive two- and four-lineage estimates still requiring a proof. They are not needed for the direct bound in Lemma S13. The current-cloud oscillation result of Karjalainen et al. (2023) does not supply the joint-lineage estimate, and the first-order path-space estimates of Chan et al. (2018) do not by themselves control the required joint-time centered covariance.

Remark 6 (Fixed-lag audit). For $0 \leq k < N$, the finite telescope for the fixed-lag score is

$$\sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^N h_s \right). \quad (\text{S371})$$

The corresponding population truncation error is

$$\sum_{s=1}^{N-k-1} \sum_{u=s+k+1}^N \Delta_{u,u-s}. \quad (\text{S372})$$

Lemma S10 and the triangle inequality therefore give

$$\|\text{Bias}_k^{\text{pop}}\|_2 \leq C(N-k-1)\sqrt{p}G'(\theta)\frac{\varrho^{k+1}}{1-\varrho}. \quad (\text{S373})$$

The sum is empty, and the population truncation bias is exactly zero, at $k = N-1$.

The former fixed-lag particle-variance proof is not valid: it applied a current-filter L^4 estimate to the historical quantity $m(\xi_{s+k})(h_s)$ with a constant independent of its genealogical age. In the constant-weight model used in Lemma S13, direct use of Equation (S337) gives instead

$$\begin{aligned} & \text{Var} \left\{ \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^N h_s \right) \right\} \\ &= \sigma^2 \left[(N-k-1)\{1 - (1-J^{-1})^{k+2}\} + \sum_{a=0}^k \{1 - (1-J^{-1})^{a+2}\} \right]. \end{aligned} \quad (\text{S374})$$

For $1 \ll k \ll \min(N, J)$ this has order $N(k+1)/J$; at $k = N-1$ it has order N^2/J before coalescence saturation. This exact calculation disproves the earlier $\{N + (k+1)^2\}/J$ upper bound. A general upper bound of order $N(k+1)/J$ requires the fixed-lag version of Equation (S367) and is not asserted here.

The same distinction applies to particle bias. The active proof of Lemma S11 establishes a uniform J^{-1} bias for a current filtering functional, but its one-sided forward-semigroup argument does not establish the same bound for the historical h_s in Equation (S371). Generic path-space weak-error results give the conservative full-score order $N(k+1)/J$. A uniform N/J result may still follow from a two-sided backward-smoothing argument, but it is not used in the verified theorem above.

Remark 7 (Why geometric discounting does not have fixed-lag bias). The exact finite-horizon scalar sum in the upper bound of Theorem S5 is

$$\sum_{r=1}^{N-1} (N-r)(1-\alpha^r)\varrho^r. \quad (\text{S375})$$

The simpler bound in Lemma S14 uses

$$\sum_{r=1}^{\infty} (1-\alpha^r)\varrho^r = \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \quad (\text{S376})$$

This identity already contains the location-forgetting factor ϱ^r . Its order in $1 - \alpha$ is not caused by discarding that factor.

Indeed, for $0 \leq \alpha < 1$ and every $r \geq 1$,

$$\begin{aligned}\alpha^r &= \sum_{k=r}^{\infty} (1 - \alpha) \alpha^k, \\ 1 - \alpha^r &= \sum_{k=0}^{r-1} (1 - \alpha) \alpha^k.\end{aligned}\tag{S377}$$

Therefore, by Tonelli's theorem for nonnegative series,

$$\begin{aligned}\sum_{r=1}^{\infty} (1 - \alpha^r) \varrho^r &= \sum_{r=1}^{\infty} \sum_{k=0}^{r-1} (1 - \alpha) \alpha^k \varrho^r \\ &= \sum_{k=0}^{\infty} (1 - \alpha) \alpha^k \sum_{r=k+1}^{\infty} \varrho^r \\ &= \sum_{k=0}^{\infty} (1 - \alpha) \alpha^k \frac{\varrho^{k+1}}{1 - \varrho} \\ &= \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}.\end{aligned}\tag{S378}$$

Thus exponential discounting is a geometric mixture of fixed-lag cutoffs. The mixing weights $(1 - \alpha)\alpha^k$ have mean $\alpha/(1 - \alpha)$, but the cutoff is zero with probability $1 - \alpha$. This mass on short cutoffs is why the mixture in Equation (S378) is of order $1 - \alpha$ rather than $\varrho^{\alpha/(1-\alpha)}$. Since

$$\frac{\alpha}{1 - \alpha} = \frac{1}{1 - \alpha} - 1,\tag{S379}$$

using $1/(1 - \alpha)$ instead as the effective horizon changes the proposed exponential rate only by the fixed factor ϱ .

The first-order behavior is also visible without taking norms. Since all sums are finite and

$$\frac{1 - \alpha^r}{1 - \alpha} = \sum_{q=0}^{r-1} \alpha^q \longrightarrow r \quad \text{as } \alpha \uparrow 1,\tag{S380}$$

Equation (S344) gives

$$\begin{aligned}\lim_{\alpha \uparrow 1} \frac{\nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta)}{1 - \alpha} \\ = \sum_{n=1}^N \sum_{r=1}^{n-1} r \Delta_{n,r}.\end{aligned}\tag{S381}$$

The assumptions do not force the vector on the right to vanish. For example, take $N = 2$, state space $\{0, 1\}$,

$$\Pr(X_1 = 1) = \frac{1}{2}, \quad \Pr(X_2 = X_1 \mid X_1) = \frac{3}{4},\tag{S382}$$

observe $Y_1 = Y_2 = 1$, and, in a neighborhood of $\theta = \phi = 0$, let

$$\begin{aligned}\Pr_{\theta}(Y_1 = 1 \mid X_1 = x) &= \frac{1}{2} + \theta \left(x - \frac{1}{2} \right), \\ \Pr_{\theta}(Y_2 = 1 \mid X_2 = x) &= \frac{1}{4} + \frac{x}{2}.\end{aligned}\tag{S383}$$

The probabilities of the complementary observations are one minus the displayed probabilities. On $|\theta| \leq 1/2$, all transition and measurement probabilities lie between $1/4$ and $3/4$, and the required derivatives are bounded. At $\theta = 0$,

$$\begin{aligned}h_1(x) &= 2x - 1, & h_2(x) &= 0, \\ \rho_1(h_1) &= 0, & \rho_2(h_1) &= \frac{1}{4}.\end{aligned}\tag{S384}$$

Indeed,

$$\Pr(Y_2 = 1 \mid X_1 = 1) = \frac{5}{8}, \quad \Pr(Y_2 = 1 \mid X_1 = 0) = \frac{3}{8},\tag{S385}$$

so $\Pr(X_1 = 1 \mid Y_1 = Y_2 = 1) = 5/8$. Consequently,

$$\nabla_{\theta}\ell(0) - \nabla_{\theta}\ell^{\alpha}(0) = (1 - \alpha)\Delta_{2,1} = \frac{1 - \alpha}{4}.\tag{S386}$$

Hence no bound of order $C\rho^{\alpha/(1-\alpha)}$, with C independent of α , can hold for all α sufficiently close to one under the present assumptions. Particular models may nevertheless have a vanishing first-order coefficient in Equation (S381).

The contraction of the location-log-weight-gradient pair does not change this conclusion. It compares two initial conditions under the same discounted recursion. By contrast, write w_n^{α} and w_n^1 only in the next display for the gradients of the corresponding log weights, evaluated along the same location path. Comparing the discounted and undiscounted recursions gives at every time

$$w_n^{\alpha} - w_n^1 = \alpha(w_{n-1}^{\alpha} - w_{n-1}^1) - (1 - \alpha)w_{n-1}^1,\tag{S387}$$

so the discount introduces a new forcing term at every step.

The finite-polynomial structure makes the necessary cancellation precise. The binomial theorem and the exact signed full-score representation give

$$\begin{aligned}\nabla_{\theta}\ell(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta) \\ = \sum_{q=1}^{N-1} (-1)^{q+1} (1 - \alpha)^q \sum_{n=1}^N \sum_{r=q}^{n-1} \binom{r}{q} \Delta_{n,r}.\end{aligned}\tag{S388}$$

Vanishing of the first signed sum in Equation (S388) improves the order from $1 - \alpha$ to $(1 - \alpha)^2$; successively higher polynomial orders require the corresponding successive signed sums to vanish. For every $c > 0$, $\rho^{c/(1-\alpha)} = o\{(1 - \alpha)^q\}$ for every fixed q as $\alpha \uparrow 1$. Hence, for fixed N , a bound of order $\rho^{c/(1-\alpha)}$ forces every coefficient in Equation (S388) to vanish. The left side is then identically zero as a polynomial in α . Equivalently, linear independence of the powers of α gives

$$\sum_{n=r+1}^N \Delta_{n,r} = 0, \quad 1 \leq r < N.\tag{S389}$$

Thus exponential decay in the mean geometric memory would assume exact all-order cancellation, rather than follow from stronger smoothness or mixing alone.

An exact summation-by-parts identity explains both the bound and the useful regime of the geometric rule. For each original time s , its contribution to the DMOP- α full score is

$$\begin{aligned} \rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\ = (1 - \alpha) \sum_{q=0}^{N-s-1} \alpha^q \rho_{s+q}(h_s) + \alpha^{N-s} \rho_N(h_s). \end{aligned} \quad (\text{S390})$$

Its exact-score contribution is $\rho_N(h_s)$, so their difference is

$$(1 - \alpha) \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \quad (\text{S391})$$

Thus geometric discounting averages partial smoothers with a geometrically distributed smoothing time capped at $N - s$, whereas fixed lag uses one smoothing time.

For a quantitative comparison, suppose $0 \leq \alpha < 1$, take $k = \lfloor \alpha/(1 - \alpha) \rfloor < N$, and compare the uncapped scalar upper bounds. Here k is the integer part of the mean geometric time. The ratio of the geometric scalar upper bound to the corresponding fixed-lag scalar upper bound is

$$\frac{\frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}}{\frac{\varrho^{k+1}}{1 - \varrho}} = \frac{(1 - \alpha)\varrho^{-k}}{1 - \alpha\varrho} \leq \varrho^{-k}. \quad (\text{S392})$$

The same ratio also satisfies

$$\begin{aligned} \frac{(1 - \alpha)\varrho^{-k}}{1 - \alpha\varrho} &= \sum_{q=0}^{\infty} (1 - \alpha)\alpha^q \varrho^{q-k} \\ &\geq \sum_{q=0}^k (1 - \alpha)\alpha^q = 1 - \alpha^{k+1} \\ &> 1 - \left(\frac{k+1}{k+2}\right)^{k+1} \geq \frac{1}{2}. \end{aligned} \quad (\text{S393})$$

Indeed, $k \leq \alpha/(1 - \alpha) < k + 1$ implies $\alpha < (k + 1)/(k + 2)$, and $\{(k + 1)/(k + 2)\}^{k+1} \leq 1/2$. Consequently the two uncapped scalar upper bounds are comparable up to a fixed factor whenever $k|\log \varrho|$ is bounded, namely when the discount memory spans at most a fixed number of mixing e-folding times. This gives a finite-regime explanation consistent with comparable empirical behavior. The asymptotic gap appears when $\alpha \uparrow 1$ while ϱ is fixed, so that the discount memory spans arbitrarily many mixing times.

DMOP- α as an estimating equation

The preceding MSE results condition on a fixed realized data set and compare DMOP- α with its realized exact likelihood score. That is not the only criterion relevant to parameter estimation. We now take expectation under the original POMP law at parameter θ_0 :

$$\begin{aligned} X_0 &\sim f_{X_0}(\cdot; \theta_0), \\ X_n | X_{n-1} &\sim f_{X_n|X_{n-1}}(\cdot | X_{n-1}; \theta_0), \\ Y_n | X_n &\sim f_{Y_n|X_n}(\cdot | X_n; \theta_0). \end{aligned} \tag{S394}$$

Thus “taking expectation over the observations” means integrating over the measurement density already specified by the POMP, rather than conditioning on the realized values $y_{1:N}^*$. It introduces no second observation model. We evaluate the population recursion at $\theta = \phi = \theta_0$.

Proposition 1 (Conditional unbiasedness). *Suppose the differentiation and domination conditions used in Equation (S150) hold also when integrating over the next observation. Then, for every fixed $\alpha \in [0, 1]$,*

$$\mathbb{E}_{\theta_0} \{ \nabla_{\theta} \ell_n^{\alpha}(\theta_0) | Y_{1:n-1} \} = 0. \tag{S395}$$

Consequently,

$$\mathbb{E}_{\theta_0} \left\{ \sum_{n=1}^N \nabla_{\theta} \ell_n^{\alpha}(\theta_0) \right\} = 0. \tag{S396}$$

Proof. For $1 \leq r < n$, the definitions of ρ_n and h_{n-r} give

$$\begin{aligned} \rho_n(h_{n-r}) &= \mathbb{E}_{\theta_0}(h_{n-r} | Y_{1:n}), \\ \rho_{n-1}(h_{n-r}) &= \mathbb{E}_{\theta_0}(h_{n-r} | Y_{1:n-1}). \end{aligned} \tag{S397}$$

Apply conditional expectation given $Y_{1:n-1}$ to the definition of $\Delta_{n,r}$ in Equation (S159). The tower property and Equation (S397) give

$$\begin{aligned} \mathbb{E}_{\theta_0}(\Delta_{n,r} | Y_{1:n-1}) &= \mathbb{E}_{\theta_0} [\mathbb{E}_{\theta_0}(h_{n-r} | Y_{1:n}) | Y_{1:n-1}] \\ &\quad - \mathbb{E}_{\theta_0}(h_{n-r} | Y_{1:n-1}) \\ &= \mathbb{E}_{\theta_0}(h_{n-r} | Y_{1:n-1}) - \mathbb{E}_{\theta_0}(h_{n-r} | Y_{1:n-1}) \\ &= 0. \end{aligned} \tag{S398}$$

It remains to treat $r = 0$. Conditional on $Y_{1:n-1}$, the simulated $X_n^{\theta_0}$ and its gradient, the next observation has density $f_{Y_n|X_n}(y | X_n^{\theta_0}; \theta_0)$. The derivative in Equation (S153) acts on both the parameter argument and the differentiable state argument. Differentiation under the observation integral therefore gives

$$\begin{aligned} &\mathbb{E}_{\theta_0} \left(h_n | Y_{1:n-1}, X_n^{\theta_0}, \nabla_{\theta} X_n^{\theta_0} \right) \\ &= \int f_{Y_n|X_n}(y | X_n^{\theta_0}; \theta_0) \nabla_{\theta} \log f_{Y_n|X_n}(y | X_n^{\theta}; \theta) \Big|_{\theta=\theta_0} dy \\ &= \nabla_{\theta} \int f_{Y_n|X_n}(y | X_n^{\theta}; \theta) dy \Big|_{\theta=\theta_0} \\ &= \nabla_{\theta} 1 = 0. \end{aligned} \tag{S399}$$

Taking conditional expectations of Equation (S399) given $Y_{1:n-1}$ and using the tower property yield

$$\begin{aligned}\rho_n(h_n) &= \mathbb{E}_{\theta_0}(h_n \mid Y_{1:n}), \\ \mathbb{E}_{\theta_0}(\Delta_{n,0} \mid Y_{1:n-1}) &= \mathbb{E}_{\theta_0}\{\rho_n(h_n) \mid Y_{1:n-1}\} \\ &= \mathbb{E}_{\theta_0}(h_n \mid Y_{1:n-1}) \\ &= 0.\end{aligned}\tag{S400}$$

Finally, Equation (S162) and the finiteness of the lag sum give

$$\begin{aligned}\mathbb{E}_{\theta_0}\{\nabla_{\theta}\ell_n^{\alpha}(\theta_0) \mid Y_{1:n-1}\} &= \sum_{r=0}^{n-1} \alpha^r \mathbb{E}_{\theta_0}(\Delta_{n,r} \mid Y_{1:n-1}) \\ &= 0\end{aligned}\tag{S401}$$

by Equations (S398) and (S400). This proves Equation (S395). Taking expectations and summing the finite sequence proves Equation (S396). $\square$

Proposition 2 (Data-averaged population-score discrepancy). *Suppose the conditions of Proposition 1 hold and the conditional scores are square integrable. Then*

$$\begin{aligned}\mathbb{E}_{\theta_0} \|\nabla_{\theta}\ell^{\alpha}(\theta_0) - \nabla_{\theta}\ell(\theta_0)\|_2^2 \\ = \sum_{n=1}^N \mathbb{E}_{\theta_0} \|\nabla_{\theta}\ell_n^{\alpha}(\theta_0) - \nabla_{\theta}\ell_n(\theta_0)\|_2^2.\end{aligned}\tag{S402}$$

If, in addition, the lagwise bound in Lemma S10, including its bound on $G'(\theta_0)$, holds with the same deterministic constants over the support of the random observations, then

$$\begin{aligned}\mathbb{E}_{\theta_0} \|\nabla_{\theta}\ell^{\alpha}(\theta_0) - \nabla_{\theta}\ell(\theta_0)\|_2^2 \\ \leq CNpG'(\theta_0)^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2.\end{aligned}\tag{S403}$$

Proof. Proposition 1, first at α and then at $\alpha = 1$, gives

$$\mathbb{E}_{\theta_0} [\nabla_{\theta}\ell_n^{\alpha}(\theta_0) - \nabla_{\theta}\ell_n(\theta_0) \mid Y_{1:n-1}] = 0.\tag{S404}$$

For $m < n$, the time- m score difference is measurable with respect to $Y_{1:n-1}$. Therefore the tower property and Equation (S404) give

$$\begin{aligned}\mathbb{E}_{\theta_0} \left[\{\nabla_{\theta}\ell_m^{\alpha}(\theta_0) - \nabla_{\theta}\ell_m(\theta_0)\}^{\top} \{\nabla_{\theta}\ell_n^{\alpha}(\theta_0) - \nabla_{\theta}\ell_n(\theta_0)\} \right] \\ = \mathbb{E}_{\theta_0} \left[\{\nabla_{\theta}\ell_m^{\alpha}(\theta_0) - \nabla_{\theta}\ell_m(\theta_0)\}^{\top} \mathbb{E}_{\theta_0} \{\nabla_{\theta}\ell_n^{\alpha}(\theta_0) - \nabla_{\theta}\ell_n(\theta_0) \mid Y_{1:n-1}\} \right] \\ = 0.\end{aligned}\tag{S405}$$

Expanding the squared Euclidean norm of the finite sum of conditional score differences and applying Equation (S405) proves Equation (S402). Under the additional uniformity condition, Lemma S14 gives, for every n ,

$$\|\nabla_{\theta}\ell_n^{\alpha}(\theta_0) - \nabla_{\theta}\ell_n(\theta_0)\|_2^2 \leq CpG'(\theta_0)^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2.\tag{S406}$$

Taking expectations and summing this inequality in Equation (S402) proves Equation (S403). $\square$

The uniform-observation condition in Proposition 2 is needed only for the last deterministic upper bound. The orthogonality identity requires only square integrability, and the uniform condition may be replaced by a random per-time bound whose second moment is uniformly bounded in n . The same proof applied to the fixed-lag tail gives, for $0 \leq k < N$,

$$\mathbb{E}_{\theta_0} \left\| \sum_{n=1}^N \sum_{r=k+1}^{n-1} \Delta_{n,r} \right\|_2^2 \leq CNpG'(\theta_0)^2 \left\{ \frac{\varrho^{k+1}}{1-\varrho} \right\}^2. \quad (\text{S407})$$

Thus both discrepancies accumulate linearly in N . Combining Equations (S392), (S403), and (S407) shows that their data-averaged upper bounds are comparable up to a fixed factor in the mixing-calibrated regime described above; the ratio for the squared bounds is the square of the ratio in Equations (S392)–(S393).

Proposition 1 does not say that a realized DMOP- α score equals the realized likelihood score when $\alpha < 1$. It says that, under the original POMP law in Equation (S394), the population conditional contributions form a martingale-difference estimating equation. The usual parametric rate therefore does not require $\alpha \rightarrow 1$, provided the estimating equation identifies θ_0 and satisfies the regularity conditions stated next.

Proposition 3 (Parametric inference for fixed α). *Fix $\alpha \in [0, 1]$. Suppose θ_0 is locally identified by the limiting population estimating equation, a local uniform law of large numbers holds, and there is a consistent solution $\hat{\theta}_N^\alpha$ of*

$$\sum_{n=1}^N \nabla_\theta \ell_n^\alpha(\hat{\theta}_N^\alpha) = 0, \quad (\text{S408})$$

where the reference value in the population recursion is set equal to its parameter argument. Suppose also that the following limits hold under the POMP law in Equation (S394):

$$\frac{1}{\sqrt{N}} \sum_{n=1}^N \nabla_\theta \ell_n^\alpha(\theta_0) \Rightarrow \mathcal{N}(0, B_\alpha), \quad (\text{S409})$$

$$-\frac{1}{N} \sum_{n=1}^N \nabla_\theta \{ \nabla_\theta \ell_n^\alpha(\theta) \}^\top \xrightarrow{p} A_\alpha \quad (\text{S410})$$

uniformly on a neighborhood shrinking to θ_0 , where A_α is nonsingular. Then

$$\sqrt{N}(\hat{\theta}_N^\alpha - \theta_0) \Rightarrow \mathcal{N}\left(0, A_\alpha^{-1} B_\alpha A_\alpha^{-\top}\right). \quad (\text{S411})$$

Proof. Apply the integral form of the multivariate Taylor expansion to Equation (S408):

$$\begin{aligned} 0 &= \frac{1}{\sqrt{N}} \sum_{n=1}^N \nabla_\theta \ell_n^\alpha(\theta_0) \\ &\quad + \left[\frac{1}{N} \int_0^1 \sum_{n=1}^N \nabla_\theta \{ \nabla_\theta \ell_n^\alpha(\theta_0 + s(\hat{\theta}_N^\alpha - \theta_0)) \}^\top ds \right] \sqrt{N}(\hat{\theta}_N^\alpha - \theta_0). \end{aligned} \quad (\text{S412})$$

Consistency of $\hat{\theta}_N^\alpha$ and the uniform convergence in Equation (S410) imply that the matrix in square brackets converges in probability to $-A_\alpha$. Nonsingularity and continuity of matrix inversion therefore give

$$\sqrt{N}(\hat{\theta}_N^\alpha - \theta_0) = A_\alpha^{-1} \frac{1}{\sqrt{N}} \sum_{n=1}^N \nabla_\theta \ell_n^\alpha(\theta_0) + o_p(1). \quad (\text{S413})$$

Equation (S411) follows from Equation (S409), Slutsky's theorem, and Equation (S413). $\square$

Remark 8 (Mixing-calibrated discounting and parameter inference). *The preceding results have three distinct consequences.*

First, suppose $0 \leq \alpha < 1$ and $k = \lfloor \alpha/(1-\alpha) \rfloor < N$. Equations (S392)–(S393) give

$$\frac{1}{2} < \frac{\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}}{\frac{\varrho^{k+1}}{1-\varrho}} \leq \varrho^{-k} = \exp\{k|\log \varrho|\}. \quad (\text{S414})$$

Hence, whenever $k|\log \varrho| = O(1)$, the geometric and fixed-lag scalar upper bounds are within fixed factors. Equivalently, discount memory spans only a fixed number of mixing e-folding times. The same comparison for the squared upper bounds has fixed factors obtained by squaring Equation (S414).

Second, expectation under Equation (S394) and martingale orthogonality give Equation (S403):

$$\begin{aligned} & \mathbb{E}_{\theta_0} \|\nabla_\theta \ell^\alpha(\theta_0) - \nabla_\theta \ell(\theta_0)\|_2^2 \\ & \leq CNpG'(\theta_0)^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2. \end{aligned} \quad (\text{S415})$$

Thus, as $\alpha \uparrow 1$, this contribution is $O\{N(1-\alpha)^2\}$ rather than the $O\{N^2(1-\alpha)^2\}$ fixed-observation upper bound in Theorem S5. The improvement from N^2 to N comes from the exact vanishing of all cross-time inner products in Equation (S405); it is not an additional mixing assumption.

Third, for every fixed $\alpha \in [0, 1]$, Proposition 3 gives

$$\sqrt{N}(\hat{\theta}_N^\alpha - \theta_0) \Rightarrow \mathcal{N}\left(0, A_\alpha^{-1} B_\alpha A_\alpha^{-\top}\right). \quad (\text{S416})$$

Therefore fixed α retains the parametric $N^{-1/2}$ rate. When $\alpha < 1$, the sandwich covariance may differ from the likelihood covariance at $\alpha = 1$, so the possible cost is statistical efficiency, not a slower consistency rate.

At $\alpha = 1$, the estimating equation is the likelihood score and the usual information equality may reduce the sandwich covariance. For $\alpha < 1$, the covariance in Equation (S411) records a possible efficiency loss, not a loss of the $N^{-1/2}$ parameter rate. This population estimating-equation result is independent of the unresolved finite- J genealogy problem. In particular, Proposition 3 does not assert an $O(N/J)$ finite-particle score bias or variance bound, nor first-order equivalence between the particle and population roots. Such a transfer requires uniform-in-parameter versions of the additional finite-particle estimates identified after Theorem S5.

S8 Figures for Bayesian Inference

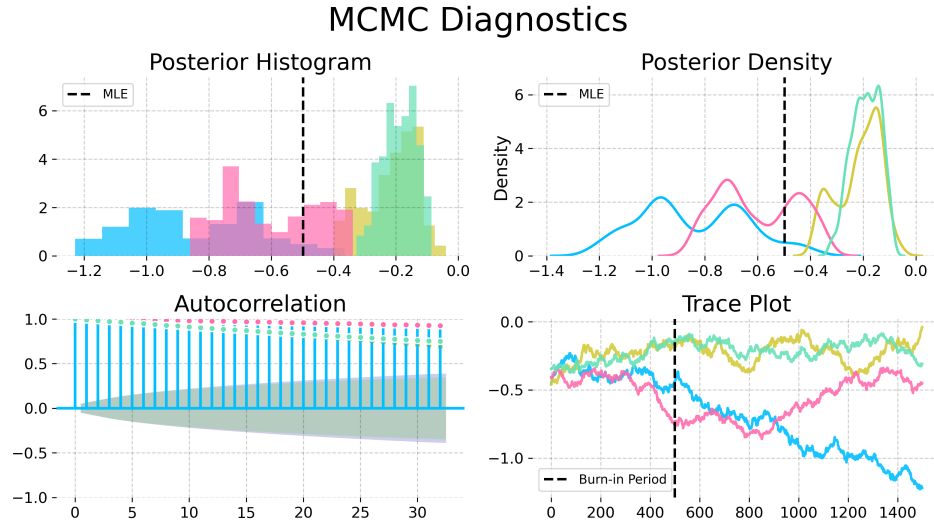


Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

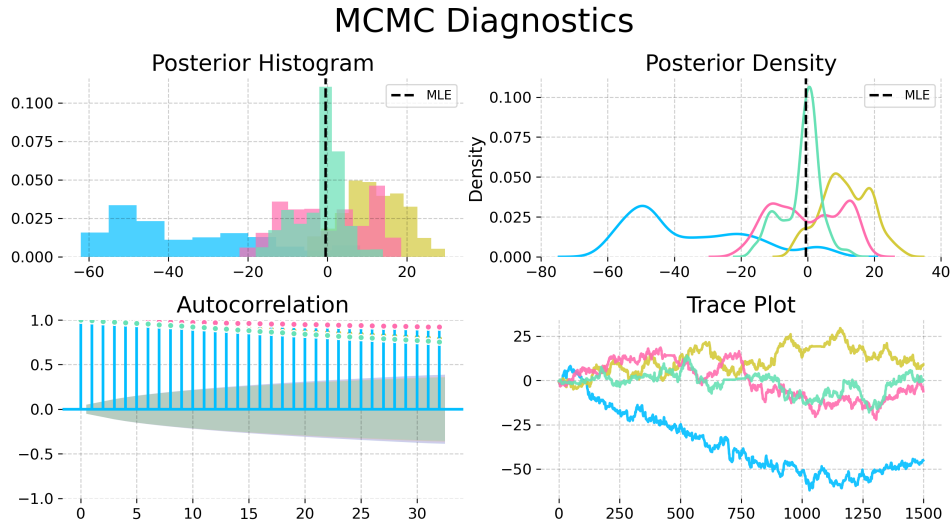


Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

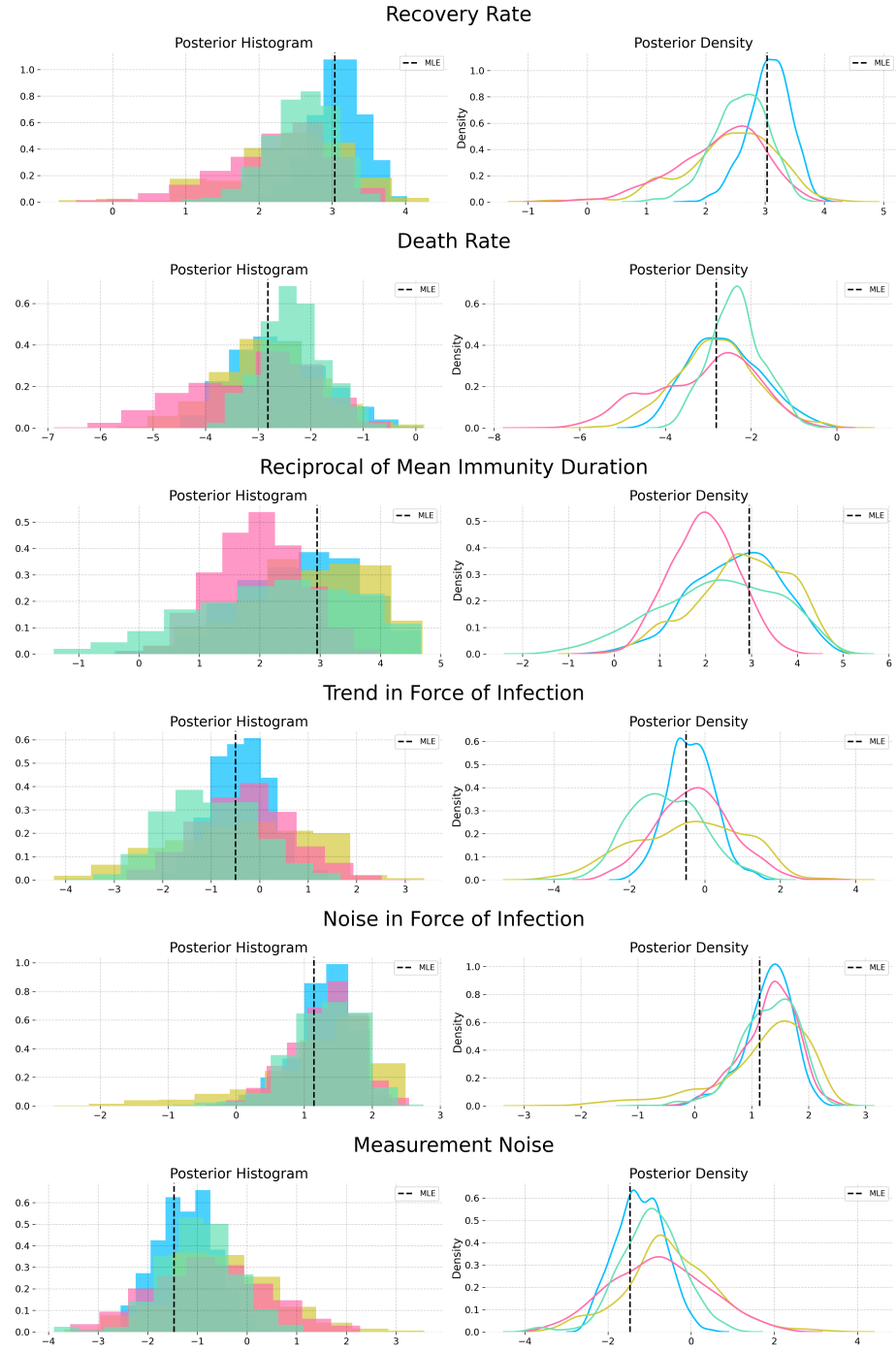


Figure S-3: Posterior estimates from NUTS powered by MOP- α with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

S9 Justification for the Choice of Baseline Comparisons

The reasoning we use in the main text to support the magnitude of our contribution is as follows: (i) iterated filtering methods are the current state of the art, and have been for a decade; (ii) we have demonstrated a substantial advance over iterated filtering. The main text focuses on demonstrating (ii), while supporting (i) via references. Listing a few references amounts only to anecdotal evidence, which is not sufficient to persuade someone properly skeptical about the claim. Therefore, this supplement provides an additional quantitative evaluation.

We count the number of papers published in three top scientific journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences of the USA (PNAS)*) that carry out statistical inference for partially observed nonlinear non-Gaussian stochastic dynamic models to address a scientific question by fitting mechanistic models to time series data. For all methods, papers in these high-profile journals are expected to represent only be the tip of the iceberg, but this count provides a concrete measure of the capability of the methods to handle world-class scientific inference problems. We could have considered additional journals, but the three we chose are undoubtedly prestigious and their scope covers the full range of scientific endeavor so there is less room for subfield bias. For iterated filtering, our high-impact application count gives the following: 1. He et al. (2023), 2. Fox et al. (2022), 3. Subramanian et al. (2021), 4. Pei et al. (2021), 5. Li et al. (2020), 6. Giezendanner et al. (2020), 7. Wale et al. (2019), 8. Pons-Salort and Grassly (2018), 9. Ranjeva et al. (2017), 10. Martinez et al. (2016), 11. Becker et al. (2016), 12. Bakker et al. (2016), 13. Laneri et al. (2015), 14. Blake et al. (2014), 15. Blackwood et al. (2013a), 16. Blackwood et al. (2013b), 17. King et al. (2008),

A referee advised consideration of an alternative approach by Gu and Kong (1998) that we found to have a high-impact application count of zero, via Google Scholar.

A referee also recommended comparison with Lindsten et al. (2014), for which the high-impact application count is also zero. Lindsten et al. (2014) has been employed once in *Nature Communications* (Calafat et al., 2018) though that journal did not quite make it onto our elite list. We also found five applications of iterated filtering in *Nature Communications* (Ranjeva et al., 2019; Rodó et al., 2021; Santos-Vega et al., 2022; Briga et al., 2024; Perez-Saez et al., 2025). There is a paper in the Statistics section of *PNAS* by Wang et al. (2022) that demonstrates the method of Lindsten et al. (2014), but here we are counting the demonstrated ability of the methodology to generate high-profile scientific papers, not statistical methodology papers. Otherwise, we would also count Ionides et al. (2006) and Ionides et al. (2015) for iterated filtering.

This reasoning supports iterated filtering as an empirically justified point of reference, in preference to the approaches of Lindsten et al. (2014) and Gu and Kong (1998). Further, the method of Lindsten et al. (2014) does not have the plug-and-play property so it cannot readily be applied to our example. The method of Gu and Kong (1998) would have to be combined with an approach such as Andrieu et al. (2010) that has known scalability problems. Therefore, we focus on comparing with iterated filtering and we do not provide a comparison with Lindsten et al. (2014) or Gu and Kong (1998).

Benchmark tasks are invaluable for making progress on challenging problems (Donoho, 2017). The Dhaka cholera data and model have served as such a benchmark since its introduction by King et al. (2008). Other papers testing methods on this example include Ionides et al. (2015); Fasiolo et al. (2016); Wycoff et al. (2026). Thus, our test is established as a demanding inference problem that challenges the best available methods. We note that this benchmark challenge contains various difficulties that arise commonly in scientific inquiry but are absent from simple SIR models such as

the simulation study considered by Lindsten et al. (2014). Ability to accommodate such features is critical for the methodology to be effective for scientific inquiry.

Iterated filtering has recently been used as a benchmark for new POMP model methodology by Whitehouse et al. (2023). This analysis presented numerical results that claimed to beat iterated filtering, but was subsequently found to contain numerical errors. Corrected results are favorable for iterated filtering methods (Hao et al., 2024; Whitehouse et al., 2025; He et al., 2025).

Recent results of Chen et al. (2025) have extended the previous theoretical understanding of iterated filtering. The ongoing study of iterated filtering as a state-of-the-art approach to inference for POMP models gives additional support for our decision to use iterated filtering as a benchmark for our new methodology.

S10 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0 , are treated differently: they are perturbed only at time t_0 , with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	Standard IF2	IFAD-0	IFAD-0.97	IFAD-1
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	650	60	175	60
GD Iterations	—	225	175	225
IF2 Cooling Rate	0.8	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemented by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2014). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not require it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

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